Managerial Allocation Skill and the Value of Digital Monitoring

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Abstract

Digital monitoring gives managers richer information, but its value depends on how managers deploy human capital. Using the staggered rollout of a tracking system in a large manufacturer, we show that monitoring raises total output by expanding span of control but does not improve average worker productivity. Productivity gains appear in teams led by managers with stronger task-allocation skill, who use the new information to deploy and retain high-performing workers more effectively. Monitoring therefore increases the returns to managerial resource-orchestration capability rather than substituting for it.

Keywords: Digital monitoring; managerial capabilities; task allocation; strategic human capital; information technology

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INTRODUCTION

Digital monitoring technologies are changing how firms see and manage work. Sensors, dashboards, barcode scans, and task-tracking systems increasingly give managers real-time information about task progress, completion times, and worker-specific performance. What managers once observed only intermittently can now be seen continuously, often at the level of individual workers and tasks. This shift matters because many frontline operational decisions are allocation decisions: managers must decide which workers should perform which jobs and how work should be distributed as conditions evolve. Digital monitoring therefore changes the information environment in which firms deploy human capital, but it does not by itself determine whether that human capital is used well.

Existing evidence explains why monitoring can improve or harm performance. Monitoring can expand managerial span of control, strengthen accountability, improve procedural compliance, and reduce moral hazard (Bloom et al., 2014; Pierce et al., 2015; Staats et al., 2017; Kelley et al., 2024; Ko and Baek, 2025). It can also backfire by crowding out intrinsic motivation, weakening cooperation, or encouraging narrow responses to measured targets (Frey, 1993; Falk and Kosfeld, 2006; Blader et al., 2020; Ranganathan and Benson, 2020). Less well understood is why similar monitoring systems generate heterogeneous effects across organizational units. We argue that a missing part of the explanation lies in the managerial capability that converts monitoring-generated information into worker–task assignments.

Specifically, we argue that an overlooked and important margin through which firms realize value from digital monitoring technologies is managerial task-allocation capability. Digital monitoring expands managers' access to information, but information is not the same as orchestration capability. Managers must decide which workers should perform which tasks, how heavily valuable workers should be loaded, and when short-run output gains risk damaging longer-run team capacity. When managers can transform monitoring-generated information into effective task assignments, monitoring can improve performance. When they

cannot, the same information may increase pressure on valuable workers, weaken retention, and leave average productivity unchanged or worse.

Our argument therefore shifts attention from the narrow question of whether monitoring "works" to why and when it creates value. We study how an information-expanding technology interacts with a managerial resource-orchestration capability. The capability is task allocation: the ability to deploy workers across heterogeneous tasks in ways that use their comparative advantages while preserving the team's future human capital. This is a central managerial problem in many organizations, especially where firms depend not only on acquiring talent but also on deploying and retaining it.

We test the argument in a large Chinese manufacturer of electrical equipment. Beginning in 2020, the firm gradually introduced a digital tracking system across production lines. The system decomposes workflows into standardized tasks, records task starts and completions in real time, and gives team managers continuous visibility into task progress and worker assignments. The rollout was staggered across product lines and, based on our balance and timing tests, was not predicted by observable characteristics of teams, workers, or managers. The setting therefore provides quasi-experimental variation in managers' access to real-time information about task execution at the worker level.

The setting is also well suited to examining task allocation as the managerial capability through which this information may create value. Team managers have limited discretion over product design or production technology, but substantial discretion over which workers perform which tasks and how workload is distributed within teams. The firm evaluates team managers on their ability to translate production targets into task assignments, match workers to jobs, and oversee execution. We use the leader allowance that senior managers assign before tracking adoption to classify high-performing managers. We then validate that this measure captures task-allocation skill using post-adoption task records: high-performing managers allocate high-performing workers away from menial tasks while leaving complex-task assignments broadly similar across manager types. This pattern is what one would

expect if stronger managers use scarce high-skill labor more selectively.

We report three main results. First, monitoring increases total output by expanding managerial span of control, but it does not improve average worker productivity. After adoption, total team output rises by 32.6 percent, team size rises by 30.3 percent, and the number of workers per manager rises by 14.7 percent, while output per worker is statistically unchanged in the average team. Second, this aggregate pattern masks sharp heterogeneity across managers. Output per worker rises in teams led by high-performing managers but falls in other teams, even though both types of teams expand. Third, several pieces of evidence support one integrated mechanism centered on task allocation. Post-adoption task records show that stronger managers deploy high-performing workers more selectively, especially by shifting them away from menial work. Those teams also retain high-performing workers after adoption, whereas other teams lose them. Survey, interview, and post-adoption worker-outcome evidence points in the same direction: workers under stronger managers report less workload stress and exhibit fewer downstream signs of overload.

We develop a simple theoretical framework to explain how these patterns may arise and when similar effects should generalize. The framework has three elements: tasks differ in how much they reward worker skill, workers differ in ability, and managers differ in their capability for using information while internalizing the retention costs of overloading valuable workers. Although managers may already have imperfect knowledge of some workers' abilities before adoption, digital monitoring makes worker-specific performance information more precise and timely. The key change is that allocation decisions become more scalable and more consequential. Stronger managers use the added information to protect scarce high-performing workers and assign them where their comparative advantage is largest. Weaker managers may use the same information to concentrate work on those workers, increasing short-run load but damaging future team composition. The framework therefore predicts heterogeneous productivity and retention effects across manager types after monitoring adoption.

Contributions. The paper contributes first to research on information technologies and organizational design. Prior work shows that information technologies can alter the span of control and structure of firms (Bloom et al., 2014). We show that the organizational value of such technologies depends on the managerial capability needed to act on the information they create. This also connects to work showing that monitoring effects depend on managerial responses: Adhvaryu et al. (2026) emphasize how managers use digitized information to target training, while Kala and Lyons (2025) emphasize how managers communicate surveillance to workers. We identify a different margin, task allocation, and show that it shapes both productivity and retention.

Second, the paper contributes to research on managerial capabilities and resource orchestration. Dynamic managerial capabilities shape how managers sense, interpret, and act on strategic information (Adner and Helfat, 2003; Helfat and Martin, 2015). Resource orchestration research similarly emphasizes that value depends on how managers structure, bundle, and deploy resources (Sirmon et al., 2007, 2011; Chadwick et al., 2015). Our contribution is not that monitoring produces information. It is that the value of this information depends on a concrete orchestration capability: deploying human capital across tasks after an information shock.

Third, the paper contributes to strategic human capital research by shifting attention from talent acquisition to talent deployment and retention. The same high-performing workers can create different value depending on how managers assign their work and whether they remain in the team. Our evidence shows that monitoring increases the returns to managers who can deploy scarce human capital without exhausting it. This complements work on managers' allocation decisions inside firms (Minni, 2026) and research on how managerial practices shape retention (Hoffman and Stanton, 2024; Benson and Shaw, 2025).

Finally, the paper is related to but distinct from Chen et al. (2026), which uses the same empirical context to study organizational learning. That paper examines how workers and managers learn different production capabilities over time, using post-tracking operational

data at the product and project level. We ask when a digital monitoring technology improves team performance, use the staggered rollout for identification, and study team managers as allocators of people and human capital. The two papers therefore differ in research question, theoretical mechanism, identification strategy, and unit of analysis.

The remainder of the paper proceeds as follows. We first describe the empirical context, data, and research design. We then develop the theoretical framework and present the main results, including the average and heterogeneous effects of monitoring. The final sections examine mechanisms, discuss managerial implications, and conclude.

EMPIRICAL CONTEXT

Our partner institution is a private Chinese manufacturer of electrical equipment, founded around 20 years ago and now a leading firm in its sector. As of 2024, it employed approximately 1,800 people and generated nearly 4 billion Chinese Yuan (¥) in annual revenue. Production is organized around product-specific lines, each comprising one to three task teams. Each team operates on a single line, is led by a manager, and includes multiple frontline workers. Our study covers 23 product lines and 38 teams. Below, we describe the institutional features most relevant for our analysis, which together make the firm particularly well suited to studying how new information technologies interact with managerial skill at the frontline level.

Team Managers. In our setting, team managers perform a focused set of frontline allocation responsibilities. After receiving daily production targets from upper management, they translate those targets into task plans, assign tasks to workers, match workers to jobs, and oversee execution. They also conduct monthly performance evaluations and distribute performance-based bonuses within their teams. All team managers are promoted internally from frontline positions; further promotion beyond this role is rare, primarily because higher-level positions require additional educational qualifications.

Although the set of tasks required for each product model is largely determined by engineering design and fixed in advance, managers retain substantial discretion over how these tasks are assigned across workers. In practice, they decide which workers are matched to which tasks and how workload is distributed within the team. Through these task-allocation choices, managers continuously shape how effectively production capacity is utilized. We use this allocation role as the relevant managerial margin throughout the paper.

Team managers receive a monthly "leader allowance," typically set at ¥300. Senior managers may adjust this amount based on their assessments of team managers' performance. These assessments are informed by regular meetings and routine shop-floor walkthroughs, and reflect senior managers' direct observations of team managers performing core frontline responsibilities, such as matching workers to jobs, distributing workload, and coordinating execution. The allowance therefore serves as an incentive tied to managerial effectiveness in deploying workers, rather than a mechanical function of short-run output targets or system-generated metrics.

Wage Determination. Workers' monthly compensation reflects their performance within the team through a two-step process. Senior management first allocates a remuneration pool to each team based on team size and overall output. This pool is then distributed to team members at the discretion of team managers. According to the firm, the amount that each worker ultimately receives depends primarily on two factors. The first is the worker's designated skill tier—Apprentice, Junior, Intermediate, or Senior—which can only be advanced through an annual formal skills assessment. The second is the monthly performance evaluation conducted by the supervising team manager.[1]

Mobility and Retention. Worker transfers across teams are infrequent and almost always initiated by workers. Interview evidence highlights workload dissatisfaction as the main driver of transfer requests, often tied to perceptions of unfair task distribution or imbalances

[1]The online appendix provides additional details on the firm's wage policy, including the determination of team-level remuneration pools and individual wages.

between effort and compensation. Other cited reasons include interpersonal tensions or skill-role mismatches. When deciding whether to request a transfer, workers typically have only limited information about alternative teams. Requests must be submitted through the current team manager and approved by upper management. In practice, most requests are granted—likely because denied requests often lead to exit—and approvals are not systematically related to workers' relative performance in their original teams.

Worker turnover is remarkably low, driven by strong community ties and the firm's reputation as a desirable local employer. Formal dismissals are rare, in part due to the rigid employment structure the firm inherited from its legacy as a state-owned enterprise.

Digital Tracking System. Beginning in 2020, the firm gradually rolled out a digital tracking system to make task progress and assignment execution more transparent. When introduced to a product line, workflows are decomposed into standardized tasks, each with a predefined expected completion time. Team managers assign these tasks via the system interface, and workers scan a QR code (or barcode) to log both the start and end times. These real-time records allow managers to observe which workers are assigned to which tasks, whether tasks have started, and whether assigned work has been completed.[2] By the end of 2023, 20 of the firm's 23 production lines (covering 92% of teams) had adopted the system. Figure 1 illustrates the staggered rollout, which, according to the firm, prioritized lines with more standardized workflows to facilitate faster and more cost-effective implementation.

DATA AND EMPIRICAL STRATEGY

We draw on a rich set of data from our partner firm, including administrative records, production logs, employee surveys, and interviews. This section describes these sources, summarizes key sample statistics, explains how we classify managerial and worker quality, and lays out the empirical strategy used to estimate the effects of digital tracking.

[2] As one manager noted in the interview, "*Previously, the supervisor had to manually inspect production progress, quality, and materials; now, all information is available at a click.*"

Data Sources

Administrative Data. Our primary administrative panel covers 1,788 individuals (team managers and frontline workers) who were employed at any time between January 2020 and September 2024, yielding 33,257 individual-month observations. It includes rich information about demographics (age, gender, education, ethnicity, and political party membership), employment details (hire date, position, team affiliation), and compensation (base pay, bonuses, and leader allowances). Subjective performance ratings assigned by team managers are available from January 2023 onward. The data also record team transfers, promotions from frontline roles to team manager positions, and transitions into non-production units.[3]

To study team-level outcomes, we aggregate the data to the team-bimonthly level. For each team in each two-month period, we construct measures of team size, managerial span of control, total output, output per worker, and turnover (entries, exits, and transfers). Direct physical output is not available in a comparable team-level measure because teams produce heterogeneous products and components. We therefore proxy total output with the remuneration pool allocated to the team before individual distribution. This pool is tied to production outcomes: senior management maps completed production, using value-based or piece-rate rules, into the team-level compensation pool from which workers are paid. The measure therefore aggregates heterogeneous production into a common monetary metric and rises when a team completes more assigned production.[4] Output per worker divides this team-level pool by team size. We use a bimonthly frequency rather than monthly to smooth fluctuations caused by volatile demand.[5]

Table 1 reports sample averages of the main variables used in our analysis. The average team consists of 16.9 members, including 1.8 managers, with each manager supervising an

[3]We do not have personnel information on the firm's top management.

[4]External demand affects the volume of work assigned to a team, but the remuneration pool remains linked to realized production because it is determined by completed output rather than by headcount alone.

[5]According to the firm, it is common for teams to receive no orders in one month and face a surge in the next, so monthly output may be an unreliable productivity measure. Bimonthly aggregation helps average out this transitory noise and mitigates the risk of bias from treatment events coinciding with demand shocks.

average of 8.75 workers. Average bimonthly team output is ¥190,230 (roughly $26,800), or ¥10,930 ($1,518) per worker. Permanent exits are rare; each team records about 1.45 internal transfers and 1.33 new hires per period. Additional demographic characteristics of workers and managers are summarized in Table A1. Women account for 18% of workers and 16% of managers. College education rates are similar across roles (21% for workers and 20% for managers), reflecting that most managers were promoted from frontline roles. Han ethnicity accounts for 95% of workers and 98% of managers, consistent with the national demographic. Managers tend to be older than workers (35 vs. 33 years on average), have longer tenure (8.4 vs. 4.2 years), and are more likely to be members of the Chinese Communist Party (CCP) or Communist Youth League (CYL) (40% vs. 26%).[6]

Production Records. From January 2022 through July 2024, we obtained detailed task-level logs from the firm's digital tracking system, covering 108,915 completed products and 466,412 task records. Each record identifies the task name, the worker who performed the task, the expected duration, start and end times, and indicators for disruptions such as material shortages. We organize these records into a worker-day panel and construct several measures. *Task Count* and *Total Workload* capture the number and expected duration of tasks assigned to a worker on a given day.[7] *Quality Defects* is a binary indicator equal to one if any task was reissued due to quality problems. *Non-Compliance* flags whether a worker failed to properly scan task completion. *No Material Shortage* captures whether a worker's task was completed without interruption from input shortages. Table A2 reports summary statistics for these measures.

The dataset contains 336 distinct task names, each corresponding to a specific production process for a given product type. We refer to each distinct task name as a task category. Using internal documentation and input from experienced employees, we further classify

[6]The CCP is the ruling party of China, and the CYL is its affiliated youth organization. Membership in either is partially merit-based, and often associated with greater opportunities for career advancement.

[7]We measure workload using expected rather than actual task duration because the latter may vary with workers' skill and proficiency, so the same task can imply different observed time spent across workers.

task categories as either *complex* or *menial*. This classification is motivated by the nature of performance variation in our setting. Physical effort and speed primarily matter for routine physical tasks, such as hoisting or cable laying. By contrast, complex tasks, such as wiring or spot welding, require workers to follow technical specifications accurately and exercise greater judgment. Overall, 15.4% of tasks in our sample are classified as complex and 84.6% as menial. Table A3 provides validation for this classification by showing that complex tasks are significantly more likely to involve execution errors.

Survey Data. In September 2024, we fielded an online survey of all production employees. Of 869 invited participants, 727 workers and all 110 managers completed the survey, which took about 15 minutes.[8] Although the survey was not anonymous, we assured respondents that all information they provided would be used only for research purposes and would be kept strictly confidential from the firm and any other organizations.

The survey asked workers about their job satisfaction and perceptions of job meaning, managerial competence, and communication effectiveness. Managers answered parallel questions about their own experience and also rated the importance of various managerial skills alongside self-assessments. We use these data primarily to explore mechanisms, particularly how variation in managerial quality relates to the effects of tracking. Key variables are summarized in Table A4.

Employee Interviews. To complement the structured data, we conducted semi-structured, anonymous interviews in April 2025 with selected frontline workers, team managers, and HR personnel, based on HR recommendations. These interviews explored topics such as transfer policies, task distribution, production disruptions, and experiences with the tracking system. They provide institutional context and help us interpret the effects of managerial practices and policies that are difficult to capture in quantitative data alone.

[8]The 32 invited but non-responding cases reflect staff turnover and an outdated mailing list that still included workers who had already left the firm. The high completion rate likely reflects both the top-down implementation of the survey, with senior leaders urging participation and team managers following up with non-respondents, and employees' strong attachment to the firm (see the Empirical Context section).

Key Variables and Identification Strategy

Dependent variables. Our headline team-level dependent variables are total output, output per worker, team size, the number of managers, workers per manager, hiring, and worker mobility. As described above, total output is measured by the team-level remuneration pool allocated before individual pay is distributed. This is the appropriate observable measure in our setting because the pool is tied to completed production and provides a common monetary metric for heterogeneous products and tasks. Output per worker divides this measure by team size. These outcomes allow us to distinguish scale effects from productivity effects. To examine mechanisms, we use worker retention outcomes, post-tracking task-allocation measures, task-execution errors, absenteeism, and survey-based workload stress.

Independent variables. The main independent variable is exposure to the digital tracking system, defined at the team-period level based on the rollout date of the team's product line. The main moderator is managerial allocation skill. We proxy for this skill using pre-tracking leader allowances and define high-performing managers as those with high average allowances before tracking. Teams are classified by whether they were led by at least one high-performing manager before implementation. We also use pre-tracking within-team performance pay to identify high-performing workers, which allows us to study whether monitoring affects their deployment and retention differently. The next subsection provides details on these classifications, explains the measurement logic behind them, and discusses their validity.

Identification strategy. The empirical strategy exploits the staggered rollout of the digital tracking system across product lines. According to the firm, implementation timing was driven by technical standardization and logistical feasibility rather than by observed team performance, managerial quality, or worker composition. We assess this claim using pre-treatment balance tests, event-study pre-trends, and checks for adoption timing based on post-implementation task complexity. The main estimates include team and period fixed

effects, time-varying team controls, and standard errors clustered at the team level.

Classifying Managers and Workers

A central question in our analysis is whether and how differences in managerial quality drive the productivity effects of digital tracking. To address this question, we classify managers and workers using observable performance indicators from the pre-tracking period. The goal is to measure managerial allocation skill before the information environment changes and to identify workers who were high-performing within their teams before tracking could affect task assignments or retention.

Measuring Manager Quality

In this setting, the relevant dimension of managerial effectiveness is task-allocation skill: the ability to translate production goals and worker information into effective worker–task matches. Team managers have little discretion over product design or technology, but substantial discretion over how workers are assigned across tasks and how workload is distributed. We therefore use the term managerial quality to refer to this allocation-oriented capability unless otherwise noted.

We measure this capability using the monthly leader allowance, the firm's own performance-based assessment of team managers. Senior leaders told us that the allowance reflects how effectively team managers match workers to tasks, distribute workload, and oversee assigned jobs. These are precisely the behaviors that define allocation skill in our setting and are difficult to observe directly in administrative data. Measures based solely on team output are less suitable because teams undertake heterogeneous tasks that complicate cross-team comparisons, and because manager rotations are rare, limiting within-team variation.

In our main specification, we compute each manager's average allowance over all pre-tracking months and define *high-performing managers* as those in the top 25% of the resulting distribution (33 of 144 managers). Our main results are robust to alternative thresholds (e.g.,

top 50%) and to specifications that use a continuous measure of managerial quality.[9]

To assess whether observable characteristics predict our classification of high-performing managers, we examine three sets of variables: managers' own demographics, the composition of their teams, and the demographics of co-managers within the same team. As Figure A1 shows, while some covariates are statistically significant predictors of the high-performing manager indicator in separate bivariate regressions, they lose explanatory power once included jointly. Consistent with this pattern, a cross-validated LASSO logit selects eight covariates for our main top-25% classification ($\lambda = 0.0403$), but none is statistically significant in the corresponding joint regression; for the alternative top-50% classification, it selects no predictors ($\lambda = 0.1618$). Figure A2 confirms the limited explanatory power of these models, with adjusted R^2 values never exceeding 0.2. Taken together, these results suggest that our classification captures meaningful heterogeneity beyond what is reflected in personnel records or team composition.[10]

Classifying Teams. To prepare for the later analysis of heterogeneous effects of digital tracking across teams, we classify teams according to whether they were led by at least one high-performing manager at any point prior to implementation. Under our main definition of high-performing managers (the top 25% of the pre-tracking leader-allowance distribution), 16 of the 38 teams qualify and are hereafter referred to as *good-manager teams*.[11]

[9]We prefer a binary classification because it facilitates interpretation in event-study analyses and highlights differences between broadly high- and low-performing team managers. The continuous measure in our robustness checks is constructed at the team level. Specifically, for each team, we take the maximum pre-tracking average leader allowance across all managers who led the team prior to implementation.

[10]Survey evidence provides additional support for the validity of our classification. Managers classified as high-performing report significantly stronger beliefs about the importance of their work and higher overall job satisfaction, consistent with prior research linking meaning of work and job satisfaction to performance (e.g., Beer et al., 2022; Judge et al., 2001). In particular, they place greater importance on their work for both the company and society, report higher satisfaction with their work status, feel more grateful toward the company, and perceive better occupational fit. They also report fewer concerns about organizational clarity, fewer perceived limits to promotion opportunities, lower dissatisfaction with benefits, less work overload, fewer interpersonal workplace conflicts, and lower effort–reward imbalance. Nevertheless, we find no significant differences between high-performing managers and others in self-evaluated skills or in the perceived importance of various managerial skills. Figure A3 summarizes these findings.

[11]Managerial assignments were largely stable over the study period: none of the 22 non-good-manager teams gained a high-performing manager after rollout, and 13 of the 16 good-manager teams retained one. Although this stability already mitigates concerns that our results are driven by shifting assignments, we

We first validate this team-level classification using pre-tracking performance. If the leader-allowance measure captures meaningful managerial allocation skill, it should predict team performance before the tracking system changes managers' information environment. Using only pre-tracking data, we regress the inverse hyperbolic sine of output per worker on an indicator for whether the team is led by a high-performing manager, controlling for team and period fixed effects as well as time-varying team-level covariates, with standard errors clustered at the team level. As shown in Column (2) of Table A5, teams achieve, on average, 13.9% higher output per worker when led by a high-performing manager ($p = 0.076$). This relationship is unlikely to be mechanical. As discussed earlier, leader allowance reflects senior leaders' assessments of managerial performance rather than a direct function of contemporaneous team output, and the team-level average leader allowance is not significantly correlated with team productivity ($p = 0.88$). The evidence therefore supports the view that our classification captures meaningful variation in managerial performance rather than simply identifying teams that were already more productive.

We also assess whether teams led by higher- and lower-performing managers differ systematically before implementation. Table A6 shows no evidence of meaningful differences in pre-implementation team composition across the two groups. We are more limited with respect to task composition, since task-allocation records are only observed after implementation. However, the post-implementation data suggest that task mix is fairly stable over time and broadly similar across manager types (Table A7).

Measuring Worker Quality

We classify workers by relative pre-tracking performance within their teams. For each worker, we compute the share of pre-tracking months in which his/her performance pay exceeded the team median. Those in the top half of the resulting worker-level distribution are classified as *high performers*. This measure has clear limitations: performance bonuses are not a

also examine outcomes following manager departures directly in the heterogeneity analysis below.

pure measure of worker ability, and they may reflect task assignments, local compensation practices, or managerial discretion. For this reason, we use the measure only as a within-team ranking, not as an absolute productivity type. This approach is appropriate because teams differ in both the nature and volume of tasks they perform, and compensation practices may vary across teams. As a validation exercise, Figure A4 shows that workers identified as high performers consistently receive higher ratings from team managers.[12] Our results are robust to an alternative definition of high performers based on the share of pre-tracking months in which a worker's performance pay ranked in the top 25% of the team.[13]

Task Allocation Skill and Manager Types

The preceding subsection defines high-performing managers using pre-tracking leader allowances and shows that this classification is not well explained by observable manager or team characteristics. We next ask whether it also predicts the behavior most closely tied to our proposed mechanism: task allocation. A challenge in studying this margin is that detailed task-allocation data are only observed after the tracking system is introduced. We therefore use post-tracking task records to show that managers classified as high-performing exhibit systematically different task-allocation behavior once the system is in place. This evidence provides a behavioral validation of the managerial-quality measure and supports using the classification as a proxy for task-allocation skill in the subsequent analysis.

We focus on data from January 2022 to July 2024, when task-level allocation information is available under the tracking system. The data are organized at the worker–day level, and we estimate the following specification:

$$Y_{iht} = \alpha_0 + \beta_1 \cdot \mathrm{GM}_{ht} + \eta_i + \theta_t + \epsilon_{iht}, \tag{1}$$

[12] We do not use these ratings directly for classification, however, because they are only available from January 2023 onward, after most teams had already adopted the tracking system.

[13] Although individual performance pay could in principle reflect favoritism, managers are required to keep daily records on efficiency, safety, teamwork, and quality that are compiled monthly to justify allocations, limiting scope for systematic bias.

where Y_{iht} denotes a worker–day outcome (e.g., workload or changes in task assignment) for worker i in team h on date t, GM_{ht} indicates whether the team is led by a high-performing manager, and η_i and θ_t are worker and date fixed effects. Standard errors are clustered at the worker level. This specification compares how the same worker is allocated tasks under different manager types, holding worker-specific attributes and time shocks constant.

We find systematic differences across manager types in workload allocation. In particular, high-performing managers appear to allocate skilled workers more selectively, reducing their overall workload rather than using them more intensively across tasks. To establish this pattern, we first estimate equation (1) using the inverse hyperbolic sine transformation of *Total Workload* (defined in the Data Source subsection). We find that high-performing workers receive 53.7% less total workload, or about 829 fewer expected minutes per day, when supervised by high-performing managers than comparable workers under other managers (Table 2). In contrast, we find no statistically significant differences in workload for low-performing workers or newly hired workers.[14]

This pattern does not reflect under-utilization of skilled workers. As one manager explained, "*Good managers give complex but lighter tasks to skilled workers and physically demanding tasks to others.*" Consistent with this account, high-performing workers under high-performing managers are assigned 50.1% less menial-task workload and complete 6.258 fewer menial tasks per day (Table A9), while the volume of complex-task assignments does not differ significantly. Together, our findings suggest that high-performing managers are more effective at assigning workers to tasks where their comparative advantage is greatest.

THEORETICAL FRAMEWORK

The empirical setting gives all managers access to richer worker- and task-level information, but the results below show that access does not produce uniform productivity gains. We use a simple theoretical framework to explain why. The framework treats digital monitoring as

[14]Results are similar when workload is instead measured using realized task duration; see Table A8.

a change in managers' information environment rather than as a technology that improves performance on its own. This focus connects the analysis to resource orchestration theory, which emphasizes that resources create value when managers structure, bundle, and leverage them through deployment decisions (Sirmon et al., 2011). In our setting, the key resource is human capital, and the relevant orchestration capability is assigning workers to tasks while preserving the future composition of the team. Digital monitoring can make this orchestration more scalable by providing managers with more precise and timely information, but it does not itself deploy resources. The purpose of our model is to clarify this mechanism rather than to reproduce every institutional detail of the firm. In doing so, it also identifies predictions that travel to other settings where information technologies expand what managers can observe but leave allocation decisions in managerial hands.

Foundations

Consider a production team led by one manager. In each period, the team must complete a mass of tasks. A share λ of tasks are complex, denoted by C, and the remaining share $1-\lambda$ are menial, denoted by M. Workers are either high-skill H or low-skill L. High-skill workers create more value on both task types, but their advantage is larger on complex tasks: $\Delta_C > \Delta_M > 0$. If η_t is the share of high-skill workers in the team and c_t^H and m_t^H are the average masses of complex and menial tasks assigned to high-skill workers, team output is

$$y_t = 1 + \eta_t(c_t^H \Delta_C + m_t^H \Delta_M). \tag{2}$$

Managers choose task assignments subject to worker capacity. The dynamic tradeoff is that high-skill workers may leave if their workload is excessive. Let $w_t^H = c_t^H + m_t^H$ be the average workload of high-skill workers. The workload-related departure rate is $d_t = \tau \max\{w_t^H - \bar{w}, 0\}$, where $\bar{w} > 0$ is the sustainable workload threshold and $\tau > 0$ determines how quickly departures rise above that threshold. Departing high-skill workers are replaced

from the general labor pool, so overloading them lowers future team composition:

$$\eta_{t+1} = (1-\rho)\eta_t + \rho\mu - (1-\rho)(1-\mu)d_t\eta_t, \tag{3}$$

where ρ is the exogenous turnover rate and μ is the high-skill share among replacement workers. This law of motion captures a simple strategic human-capital problem: assigning valuable workers to more tasks can increase current output, but doing so too aggressively can reduce future human capital.

Digital monitoring changes the information environment. Before monitoring, managers may have partial knowledge about some workers, but their information is noisy, delayed, and difficult to scale. After monitoring, they observe more precise and timely worker–task performance signals. The technology therefore increases the scope for targeted assignment, but it does not determine how assignments are made.

Intuition

The value of monitoring depends on how managers use the additional information. Managers with stronger task-allocation skill recognize both sides of the allocation problem. They assign high-skill workers to complex tasks where their comparative advantage is largest, and they avoid pushing those workers beyond the sustainable workload threshold. In steady state, this preserves the stock of high-skill workers and raises productivity.

Managers with weaker task-allocation skill may respond differently. Once high-performing workers are more visible, weaker managers may rely on them too heavily, including for routine menial work where their advantage is smaller. This can raise short-run load but trigger departures among precisely the workers whose retention matters most. In that case, monitoring can expand span of control and total output while failing to raise, or even lowering, output per worker.

The framework shows how the focal mechanism can travel beyond the specific technology

and setting. The key ingredients are not unique to our partner manufacturer: tasks differ in their returns to skill, workers differ in productivity, managers allocate workers across tasks, and valuable workers can leave or transfer when workload becomes unsustainable. The same mechanism should apply in other settings where information technologies make worker performance more visible while managers differ in how effectively they use that information.

Predictions

The framework yields four empirical predictions. We state and briefly explain them here, leaving the full derivations and model details to the online appendix.

Prediction 1: Monitoring should increase scale more reliably than average productivity. Better information can expand managers' span of control and allow teams to grow, so total output should increase after adoption. Average worker productivity, however, need not increase because productivity depends on how managers use the information.[15]

Prediction 2: Productivity gains should be concentrated among teams led by managers with stronger task-allocation skill. If high-performing managers are more effective at using monitoring-generated information to improve worker–task matching, output per worker should rise more in their teams than in teams led by other managers.

Prediction 3: Retention of high-performing workers should diverge across manager types. Task allocation that keeps workload sustainable should help preserve high-performing workers. These workers should therefore be more likely to remain in teams led by high-performing managers and more likely to transfer out of other managers' teams or exit after adoption.

Prediction 4: Post-monitoring task records should reveal different allocation behavior. High-performing managers should allocate high-performing workers more selectively, especially by reducing their lower-return menial workload rather than simply using them more intensively across tasks.

[15]Because team size and task volume are fixed in the model, the total-output prediction is not formally derived in the baseline formulation. It follows from a straightforward extension in which monitoring expands the scale of work a manager can supervise. Such an extension would add a scale channel without changing the model's main implications for average productivity.

MAIN RESULTS

Guided by the theoretical predictions, this section evaluates the effects of the digital tracking system using its staggered rollout across product lines. We first document average changes in team-level outcomes following implementation. We then turn to heterogeneity in productivity responses, showing that per-capita productivity gains accrue primarily to teams led by high-performing managers, whereas other teams experience weak or negative effects.

Research Design

Our identification strategy exploits the staggered rollout of the digital tracking system across product lines, which exposed different teams to the technology at different points in time. Rollout was staggered because implementation was operationally intensive: the firm had to define task categories, develop and customize the software, train managers and workers, conduct tests, and only then formally launch the system. Although the firm reports that implementation timing was guided by technical standardization and logistical feasibility, potential endogeneity concerns remain. In particular, teams that adopted earlier may differ systematically from later adopters along dimensions that also affect performance, such as product type, task complexity, or managerial capability.

To assess the validity of our identification strategy, we pursue a set of pre-treatment checks and robustness exercises. We begin by examining the parallel trends assumption. The event-study estimates of the dynamic effects of digital tracking show that the key outcomes followed similar paths for the treated and untreated teams prior to rollout (see Figure 2 and Figure A5). We next conduct three balance checks, with details provided in the online appendix. As summarized in Table A11, the results show that, prior to implementation, neither the levels nor the trends of worker and manager demographic characteristics within teams differ significantly by whether a team ever adopts the system or by the timing of adoption.

A further concern is whether adoption timing is correlated with task composition, especially the share of complex tasks. Because pre-implementation task records are unavailable, we cannot test this directly. To probe this concern despite the data constraint, we use post-implementation task records, leveraging again the fact that task mix appears fairly stable over time at the team level. Specifically, we regress a team's implementation order on the level and first difference of its complex-task share. Table A10 shows no statistically significant relationship. We view this result as only suggestive, but it provides some reassurance that adoption timing is not systematically related to task complexity.

Result 1: Average Effects of Monitoring

To estimate dynamic effects of system exposure, we use the following two-way fixed effects (TWFE) event-study specification:

$$Y_{ht} = \alpha_0 + \sum_{k=\underline{k},\, k\neq -1}^{\overline{k}} \delta_k D_{ht}^k + \gamma' X_{ht} + \phi_h + \theta_t + \epsilon_{ht}, \tag{4}$$

where Y_{ht} is the outcome for team h in bimonthly period t, and D_{ht}^k is an event-time indicator equal to one if $t = \tau_h + k$, with τ_h being the treatment date for team h. The event window is defined by $\underline{k} = -6$ and $\overline{k} = 12$, with $k = -1$ omitted as the reference period. The vector X_{ht} includes time-varying team-level controls, and ϕ_h and θ_t denote team and period fixed effects, respectively. For continuous outcomes, we apply the inverse hyperbolic sine (IHS) transformation, which accommodates right-skewness and zero values and permits percentage-based interpretation. We estimate these models by ordinary least squares. For count variables, we use Poisson regression. Standard errors are clustered at the team level.

Figure 2 reports the estimated dynamic effects. Following adoption, total team output, team size, and span of control all rise markedly (Panels (a), (c), and (e), respectively). By contrast, output per worker displays no significant aggregate change (Panel (b)). Some pre-trend concerns are visible in the baseline TWFE event studies for total output, team size,

and span of control. However, these patterns are substantially attenuated once we apply the staggered-adoption corrections introduced below.

To further assess whether the average effects are driven by residual pre-trends in the baseline TWFE specification, we implement a simple detrending exercise. For each outcome, we group teams into cohorts defined by implementation timing and estimate cohort-specific linear trends using only pre-implementation observations. We then subtract the fitted trends from the outcome in all periods and re-estimate the baseline average-treatment-effect specifications using the detrended outcomes. Table A12 reports the results for the outcomes with the clearest pre-trend concerns noted above. The estimated effects on total output and team size remain positive and statistically significant, indicating that the scaling effects of digital tracking are not driven by linear pre-trends. For workers per manager, the detrended estimate remains positive but imprecise. This outcome, however, exhibits weaker pre-trend concerns under the staggered-adoption corrections and is not central to our main argument.

Alternative Estimators. Because treatment is staggered, we also estimate the event studies using the approaches of Callaway and Sant'Anna (2021), Sun and Abraham (2021), and De Chaisemartin and d'Haultfoeuille (2020). Figure A6 shows that the corrected estimates are consistent with the baseline patterns.

Average Treatment Effects of System Exposure

To obtain the average treatment effect, we estimate a difference-in-differences specification:

$$Y_{ht} = \alpha_0 + \beta \cdot \text{Treat}_{ht} + \gamma' X_{ht} + \phi_h + \theta_t + \epsilon_{ht}, \tag{5}$$

where Treat_{ht} is an indicator equal to one if team h has been exposed to the system in period t. Other variables and estimation choices are as in the event-study specification.

Table 3 reports the results. System exposure increased total output by 32.6%, team size by 30.3%, and managerial span of control by 17.7%. It also brought about a 78.8% increase

in the number of new hires. By contrast, we find no significant change in output per worker or in the number of managers. We also examine potential effects on worker mobility, but detect no statistically significant impacts (see Table A13).

The increase in total output aligns closely with the pattern observed in the event study, reinforcing the conclusion that the system had a positive impact on total team output. The results also suggest that a key driver of this growth was an expansion in the span of control: by enhancing the oversight capacity of existing managers, the system enabled the firm to scale up its workforce through more active hiring, without needing to increase the number of managers. This interpretation is supported by qualitative evidence from field interviews. As one manager noted, "*Managing larger teams has become easier—previously 10 workers was challenging but now teams with more than 20 workers can be managed similarly.*" However, neither the event study nor the difference-in-differences estimates reveal a significant change in average output per worker at the aggregate level. As we show next, this pattern masks substantial heterogeneity driven by differences in managerial quality.

Result 2: Heterogeneity by Managerial Allocation Skill

To assess whether the effects of system adoption vary by managerial quality, we estimate a difference-in-differences model with interactions:

$$Y_{ht} = \alpha_0 + \beta_1 \text{Treat}_{ht} + \beta_2 \text{Treat}_{ht} \times \text{Good}_h + \gamma' X_{ht} + \phi_h + \theta_t^g + \theta_t^b + \epsilon_{ht}, \tag{6}$$

where Good_h indicates whether team h is a good-manager team (i.e., it had at least one high-performing manager prior to adoption; see the Measuring Manager Quality subsection), and θ_t^g and θ_t^b are separate time fixed effects for good- and non-good-manager teams, respectively. We include these group-specific time fixed effects to align the specification with the grouped event-study design.

Table 4 reports the estimates, and Figure 3 presents the corresponding event study results.

Although total output increased significantly across all teams, the effects on per-worker productivity were markedly different: output per worker increased by 17.9% more in good-manager teams than in other teams. These contrasting outcomes reflect differences in how teams scaled following adoption. While both types of teams expanded, teams in the good-manager group did so more efficiently: team size increased by 31.7% on average, compared to 54.0% in other teams. In short, good-manager teams achieved output gains with more modest increases in headcount, translating into higher per-worker productivity.

We conduct several additional analyses to assess the robustness of the heterogeneous effects. First, the results are not sensitive to how managerial quality is classified. Defining high-performing managers as those above the median, rather than the top 25%, yields similar estimates (Figure A9 and Table A15). A fixed-effects-based classification, following Abowd et al. (1999) and Lazear et al. (2015), produces broadly consistent patterns, albeit with lower precision due to limited manager rotation. Second, the results are also robust to alternative definitions of worker quality. When high-performing workers are identified by the frequency with which their performance pay ranked in the team's top 25%, rather than above the median, the estimated effects remain similar (Table A16). Third, restricting the sample to incumbent workers yields similar differences in productivity gains, indicating that the effects are not solely driven by newly hired employees (Figure A10). Fourth, excluding teams that lost their high-performing managers after system adoption does not materially change the findings (Table A17). Moreover, relative to teams that never had a high-performing manager, the excluded teams experienced a 21.8% decline in per-worker productivity after losing a high-performing manager, controlling for team and bimonthly period fixed effects (Table A18). Finally, replacing the binary indicator Good_h with a continuous pre-tracking measure of managerial performance based on leader allowance yields qualitatively similar results (Table A19).

MECHANISMS

The main results show that monitoring raises total output on average, but productivity gains are concentrated in teams led by high-performing managers. These results are consistent with the resource-orchestration mechanism at the heart of our theoretical framework: digital tracking improves visibility, but performance improves only when managers use that visibility to allocate labor selectively and sustainably.

We have already provided behavioral validation for this allocation margin in the Task Allocation Skill and Manager Types subsection. Using post-tracking worker-day task records, we show that, compared with other managers, high-performing managers assign high-performing workers lower total workload and lower menial-task workload, while complex-task volume does not differ significantly. This evidence supports the view that high-performing managers create value from tracking not by using their strongest workers more intensively, but by deploying them more selectively.

In this section, we provide additional evidence on this mechanism by examining its downstream implications. The first is retention. In the theoretical framework, selective use of high-performing workers matters because excessive reliance on them can erode the future stock of high-performing workers in the team. We test this dynamic margin directly using retention outcomes, which are observed both before and after tracking. The second implication concerns worker experience. If high-performing managers deploy high-performing workers more sustainably, these workers should exhibit fewer downstream symptoms of overload. We test these implications using post-implementation task-execution records, absenteeism data, and survey measures of workload stress. After presenting these tests, we consider several alternative explanations and present evidence that favors our interpretation.

Retention, Composition, and Counterfactual Performance

If high-performing managers use tracking to allocate work more selectively, high-performing workers should be more likely to remain in their teams. This is what we observe. Replicating the grouped event-study and heterogeneous treatment analyses for high-performing-worker transfers, we find that high-performing workers are less likely to leave good-manager teams but more likely to leave other teams after tracking is introduced. Column (3) of Table 4 and Panel (c) of Figure 3 show a large and statistically significant interaction effect. Figure A8 shows the corresponding change in team composition: in good-manager teams, the number of high-performing workers rises after adoption while their share remains broadly stable; in other teams, the share of high-performing workers declines.

These retention patterns raise a further question: do the heterogeneous productivity effects mainly reflect changes in how incumbent workers perform, or do they also reflect changes in who remains in the team? To separate these margins, we construct a counterfactual outcome that holds each team's worker mix fixed at its pre-implementation composition while allowing average performance pay within worker groups to evolve over time. The online appendix provides the details. In brief, we first group workers into old high performers, old lower performers, and new workers. We then compute each group-specific average performance pay in each team-period and weight those averages by the team's pre-implementation group shares. The counterfactual asks how team performance pay would have evolved if the baseline worker mix had been preserved.

Table A20 reports the results. The average effect of tracking remains positive using the counterfactual outcome, but the productivity difference between good-manager and other teams becomes substantially smaller and statistically insignificant. This pattern matters for the mechanism because it shows that retention and team composition are not incidental. Monitoring increases the visibility of valuable workers, and managerial allocation capability shapes whether those workers remain in the team and continue contributing to realized productivity. The exercise is not a sharp decomposition: because it uses the outcomes

of workers who remain, it may understate productivity losses that initially induced some high-performing workers to transfer out. Still, the attenuation in Table A20 indicates that differential retention is quantitatively important.

Post-Adoption Worker Outcomes and Survey Evidence

The next step is to examine whether workers' observed experience is consistent with the retention channel. We focus on task-execution errors, absenteeism, and perceived workload stress, each tied to a specific data source. These outcomes are measured after adoption, so they are mechanism-consistent evidence rather than standalone causal tests.

First, post-adoption task records show fewer execution problems under high-performing managers. We identify task-execution errors using task reassignments: if a worker is reassigned a task previously marked as complete, the original task likely failed inspection. Panel A of Table A21 shows that high-performing workers supervised by high-performing managers are 9.4 percentage points less likely to make errors, relative to a sample mean of 0.356. The same table shows no comparable reduction for workers outside the high-performing group, while new hires in good-manager teams also have lower error rates. This pattern is consistent with the task-allocation results reported in Tables 2 and A9: the strongest worker-outcome differences appear where the task-allocation differences are most pronounced.

Second, administrative absence records point in the same direction. Panel B of Table A21 shows that high-performing workers under high-performing managers are 1.4 percentage points less likely to be absent, relative to a mean absence rate of 0.5 percent. We do not find a statistically significant difference for workers outside the high-performing group. Interviews help interpret the estimate: workers and managers describe fatigue and excessive workload as sources of mistakes, stress, and absence, and managers report watching for early signs of fatigue when assigning work.

Third, the survey evidence links the same pattern to workers' own assessments of workload pressure. Table A4 summarizes the September 2024 survey, and Table A22 focuses on

high-performing workers' responses to the statement "I feel that my workload is too heavy." Conditional on demographics and performance-based pay, high-performing workers supervised by high-performing managers are 74.1 percentage points less likely to report overload, relative to a mean of 0.688 among peers supervised by other managers. The small number of surveyed high-performing workers under high-performing managers warrants caution, but the direction and magnitude are consistent with the administrative evidence.

Taken together, the worker-outcome evidence strengthens the mechanism. Tables 2 and A9 show that high-performing managers deploy high-performing workers differently; Table 4 and Figure 3 show that those workers are differentially retained; Figure A8 shows the corresponding composition changes; and Tables A21 and A22 show that the day-to-day worker outcomes move in the expected direction. The evidence therefore points to a coherent sequence from monitoring-generated visibility to task allocation, worker experience, retention, and realized team productivity.

Alternative Explanations

We also examine alternative explanations (more details can be found in the online appendix). The heterogeneous productivity and retention patterns do not appear to be driven by differential financial rewards or promotion opportunities: high-performing workers under high-performing managers do not receive significantly higher performance pay or faster promotion after adoption (Table A23). They also do not appear to be driven by better system compliance alone. High-performing managers are modestly more successful at ensuring accurate task closure in the tracking system, but this effect is concentrated among high-performing workers and cannot account for broader team-level productivity gains (Tables A24 and A25). Finally, material shortages and broader workplace perceptions are balanced across manager types (Tables A24 and A26). Together, the evidence favors the task-allocation mechanism.

MANAGERIAL IMPLICATIONS

The findings point to a managerial lesson about complementarities between digital technologies and managerial capabilities. Digital monitoring can make work visible, expand span of control, and help firms scale output. But visibility is only an input. The organizational return depends on whether managers can convert that information into effective deployment of human capital.

This distinction changes how firms should evaluate monitoring investments. If senior leaders focus only on total output or team growth, they may interpret scale as evidence that the system is creating productivity value. Our results show that these outcomes can diverge: monitoring can allow existing managers to supervise more workers while leaving average productivity unchanged. Implementation reviews should therefore separate scale effects from deployment effects. The latter require asking whether the technology improves how workers are matched to tasks, not just whether it increases how much work is observed.

The heterogeneity results make the complementarity more concrete. Teams led by high-performing managers increase output per worker after adoption, whereas other teams do not. The same technology therefore creates different returns depending on the allocation capability of the managers who receive the information. It is not enough to have the right workers or the right data. Firms also need managerial routines and incentives that help allocate valuable workers to high-return tasks without exhausting the human capital on which future performance depends.

This has direct implications for capability development and managerial design. Firms introducing monitoring systems should treat task-allocation capability as a specific managerial skill: interpreting worker-level information, matching workers to jobs, distributing workload, and preserving valuable workers over time. Training and evaluation systems should therefore focus less on generic exhortations to "use data" and more on whether managers make selective, sustainable assignment decisions as span of control expands.

The findings also matter for strategic human capital management. Monitoring makes high-performing workers easier to identify, but visibility can also make them easier to overuse. In our data, teams with weaker managers lose high-performing workers after adoption, whereas teams with stronger managers do not. The organizational challenge is therefore not only selecting or retaining talented workers in the abstract, but designing managerial systems that deploy them without degrading their willingness to stay.

This interpretation changes what firms should monitor during rollout. The most informative post-implementation indicators may include whether high-performing workers are being concentrated in routine or physically demanding work, whether these workers are leaving particular teams, and whether workload-related strains appear in errors, absenteeism, or worker reports of overload. In our setting, those outcomes move together. They can therefore serve as warning signs that information generated by the system is being converted into excessive pressure rather than better orchestration.

More broadly, monitoring systems can widen performance differences across managers. Stronger managers use added visibility to scale while maintaining disciplined task allocation. Other managers may also scale, but in ways that dilute productivity and weaken retention. Pairing digital monitoring with managerial capability development is therefore not merely a complementary investment. It is part of the organizational design required to govern the consequences of making work more visible.

CONCLUSION

Digital monitoring expands what managers can see, but not all managers can use that information equally well. In our setting, monitoring increases total output by expanding span of control, but average productivity gains appear only where managers have stronger task-allocation skill. The broader implication is straightforward: information-expanding technologies create value when they are matched with the managerial capabilities needed to use the information well. They do not replace those capabilities.

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MAIN TABLES AND FIGURES

Table 1: Summary Statistics (Team × Bimonthly Period Level)

Variable	Obs. #	Mean	Std. Dev.	10%	25%	50%	75%	90%
Panel A. Output Variables								
Total Output (thousand yuan)	1029	190.23	157.59	52.70	84.34	137.70	244.85	411.18
Output per Worker (thousand yuan)	1029	10.93	3.05	7.09	8.79	10.86	12.89	14.71
Panel B. Personnel Variables								
Team Size	1029	16.93	11.69	5	8	13	23	35
Worker per Manager	1029	8.75	5.62	3.41	5.00	7.50	11.00	15.00
No. of Managers	1029	1.84	1.18	1	1	1	2	3
No. of New Workers	1029	1.33	2.57	0	0	0	2	4
No. of High Performers Transferred	1029	0.93	2.52	0	0	0	0	3
No. of Low Performers Transferred	1029	0.52	1.48	0	0	0	0	2
Turnover Rate for High Performers	1029	0.00	0.02	0	0	0	0	0
Turnover Rate for Low Performers	1029	0.01	0.03	0	0	0	0	0
Panel C. Managerial Quality Variable								
Team with High-Performing Managers	1029	0.41	0.49	0	0	0	1	1

Notes. This table summarizes the team-level bimonthly panel dataset. Panel A reports output measures. Total output (in thousand Chinese Yuan) is defined as the total wage paid to all team members in a given period. Output per worker (in thousand Chinese Yuan) is calculated as total output divided by the number of workers. Panel B presents human resource variables. Team size refers to the total number of employees assigned to the team, including both managers and frontline workers. Worker per manager captures managerial span of control and is defined as the ratio of workers to managers within a team. The number of new workers refers to externally hired employees who joined the firm in a given period and were allocated to the team. High and low performer transfers capture the number of workers of each type reassigned across teams within the firm. Turnover rates are calculated as the number of workers leaving the firm in a given period divided by team size, and are reported separately for high and low performers. Panel C includes the indicator for whether the team is led by a high-performing manager.

Table 2: Differences in Workload Allocation across Manager Types

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	IHS(Workload)				Workload			
	All Workers	High Performers	Low Performers	New Workers	All Workers	High Performers	Low Performers	New Workers
Supervised by a High-performing Manager	-0.347	-0.537	0.098	1.405	-593.189	-829.465	0.810	1,016.306
	(0.123)	(0.129)	(0.279)	(0.783)	(228.461)	(291.060)	(85.453)	(631.722)
Observations	12,962	8,169	2,748	1,920	12,962	8,169	2,748	1,920
R-squared	0.522	0.526	0.518	0.687	0.352	0.343	0.473	0.497
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. This table reports the difference in task allocation styles across different manager types (high-performing vs. others). The key independent variable is a time-varying indicator for whether the worker was supervised by a high-performing manager. The dependent variable of Columns (1)–(4) is the IHS-transformed value of worker's daily assigned workload, and that of Columns (5)-(8) is the raw value of worker's daily assigned workload. The unit of observation is a worker-day. All columns are estimated using OLS. Columns (1) and (5) include all workers; Columns (2) and (6) focus on high performers; Columns (3) and (7) on low performers; and Columns (4) and (8) on new hires who joined after the system implementation. All regressions include worker and date fixed effects and cluster standard errors at the worker level.

Table 3: Average Treatment Effects on Main Team-Level Outcomes

Variable	(1) IHS(Output)	(2) IHS(Output per Worker)	(3) Team Size	(4) No. of Managers	(5) IHS(Worker per Manager)	(6) No. of New Workers
Implemented	0.326	0.042	0.265	0.164	0.177	0.581
	(0.125)	(0.036)	(0.126)	(0.142)	(0.075)	(0.181)
Observations	955	955	955	955	951	910
R-squared	0.810	0.679			0.541	
Control Mean	12.36	9.83	17.01	2.03	2.73	1.52
Team Controls	Yes	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. This table reports estimates of the average treatment effect of the digital tracking system on key team-level outcomes. The treatment indicator equals one if the product line associated with a team had implemented the system during the period. The unit of observation is a team-bimonthly spell, and the sample covers the period from January 2020 to September 2024. Columns (1), (2), and (5) are estimated using OLS, while columns (3), (4), and (6) use Poisson regressions given the count nature of the dependent variables. In the Poisson specifications, team-period observations with all-zero outcomes are dropped, leading to smaller sample sizes. All regressions include team and period fixed effects, along with time-varying team-level covariates: the share of workers who are female, ethnic Han, and college-educated, as well as the average age and tenure of both workers and managers. Control Mean reports the average of the dependent variable among untreated observations. Standard errors are clustered at the team level. Results for all available dependent variables are presented in Table A13.

Table 4: Heterogeneous Effects by Managerial Quality

Variable	(1) IHS(Output)	(2) IHS(Output per Worker)	(3) High Performers Transferred Out
Implemented	0.367	-0.063	1.040
	(0.121)	(0.030)	(0.632)
Implemented × Good-Manager Team	-0.018	0.179	-1.713
	(0.212)	(0.066)	(0.816)
Observations	955	955	642
R-squared	0.819	0.697	
Control Mean	12.28	9.83	0.42
Team Controls	Yes	Yes	Yes
Team FE	Yes	Yes	Yes
Group × Period FE	Yes	Yes	Yes

Notes. This table reports estimates of the effect of the digital tracking system and its interaction with managerial quality on key team-level outcomes. Managerial quality is based on the average pre-tracking leader allowance. The good-manager team indicator equals one if the team was ever led by a manager ranked in the top 25% of this distribution before treatment. The treatment indicator equals one if the product line associated with a team had implemented the system during the period. The managerial quality indicator equals one if the team had at least one good manager at any point prior to being exposed to the system. The unit of observation is a team-bimonthly spell, and the sample covers the period from January 2020 to September 2024. Columns (1) and (2) are estimated using OLS, while column (3) uses a Poisson regression given the count nature of the dependent variable. In the Poisson specification, team-period observations with all-zero outcomes are dropped, leading to a smaller sample size. All regressions include team and period fixed effects, along with time-varying team-level covariates: the shares of workers who are female, ethnic Han, and college-educated, as well as the average age and tenure of both workers and managers. Control Mean reports the average of the dependent variable among untreated observations. Standard errors are clustered at the team level. Results for all available dependent variables are presented in Table A14.

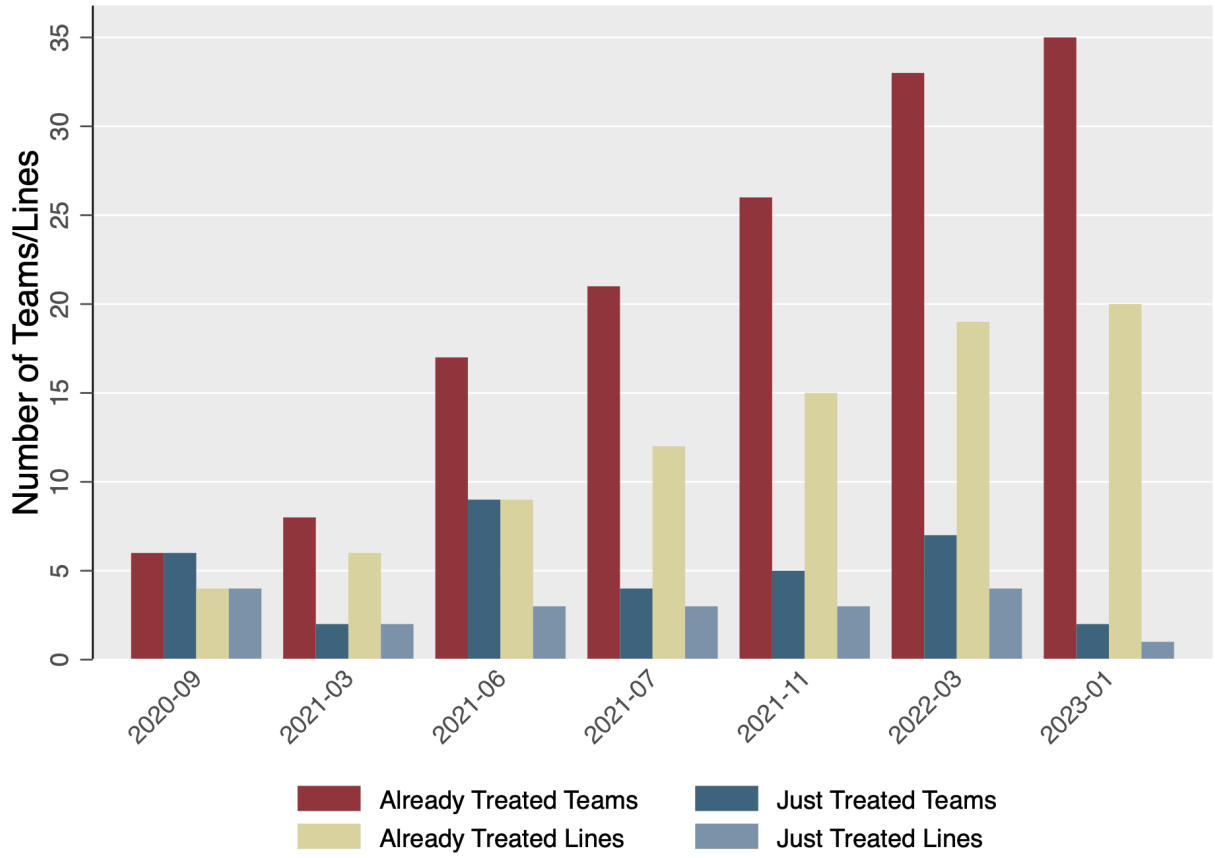
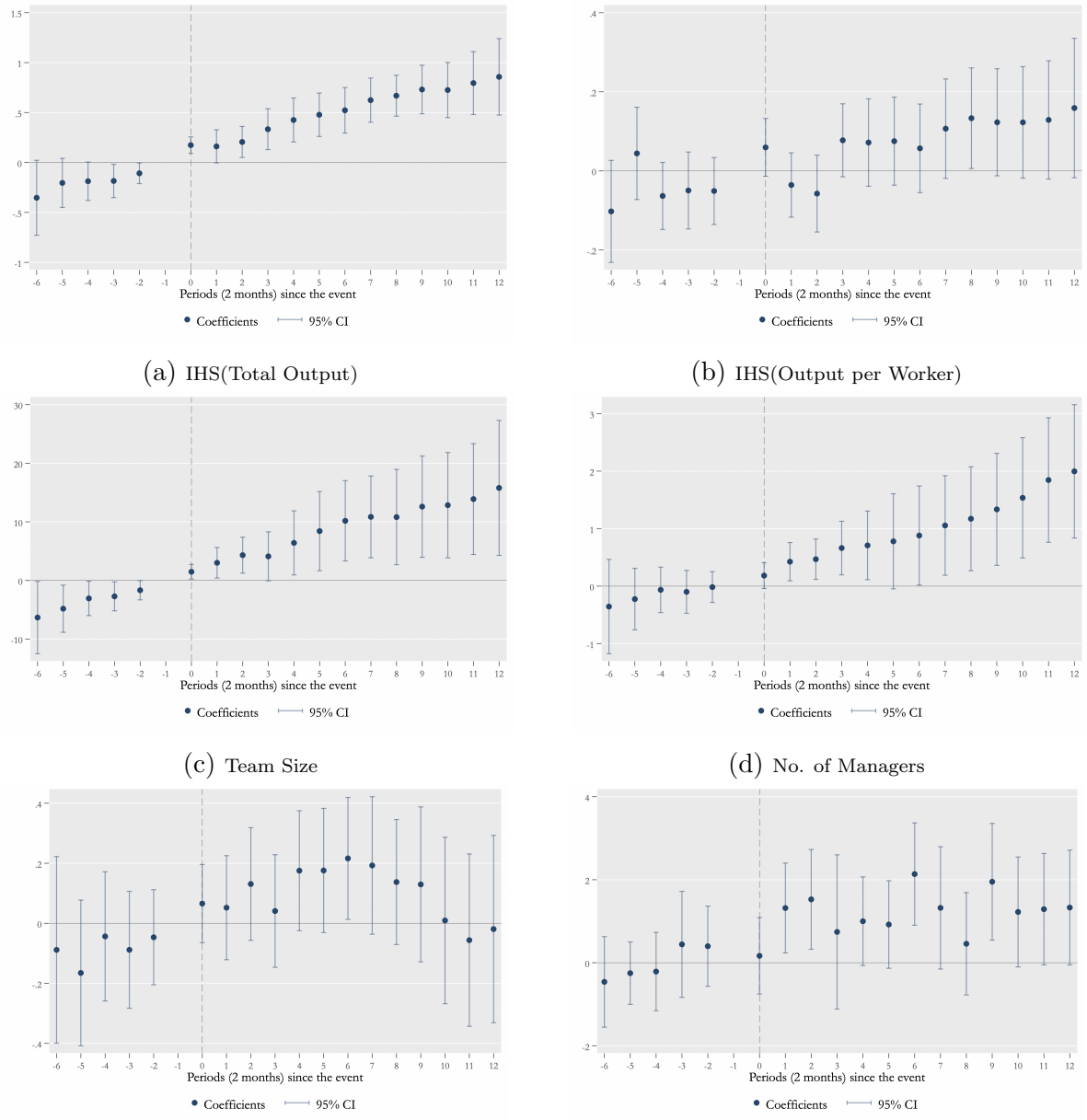


Figure 1: Rollout of the Digital Tracking System

Notes. This figure presents the staggered rollout of the digital tracking system across teams and product lines over seven implementation waves between September 2020 and September 2023. The yellow bars indicate the cumulative number of product lines that had adopted the system by each wave, while the light blue bars indicate the number of new product-line adoptions in each wave. The dark red bars indicate the cumulative number of teams exposed to the system, and the dark blue bars indicate the number of newly exposed teams. Three product lines, each comprising a single team, never adopted the system.



(a) IHS(Total Output)

(b) IHS(Output per Worker)

(c) Team Size

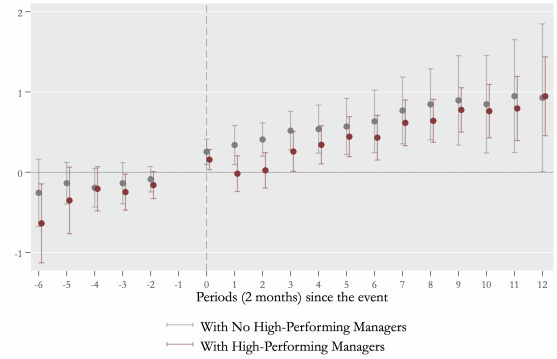
(d) No. of Managers

(e) IHS(Workers per Manager)

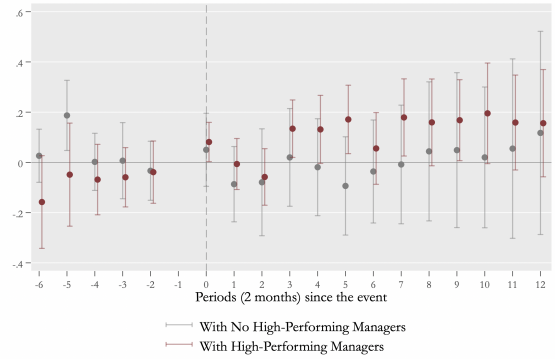
(f) No. of New Workers

Figure 2: Team-level Event Studies

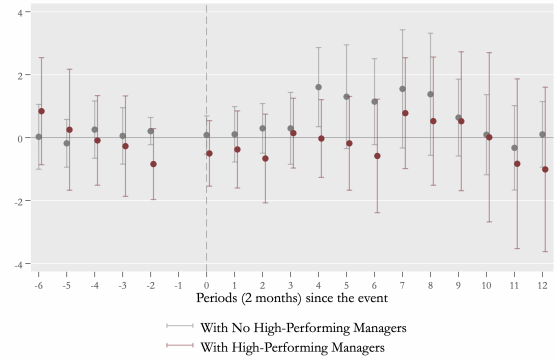
Notes. This figure presents event-study estimates and 95 percent confidence intervals from the estimation of equation (4), with each panel corresponding to a different team-level outcome. The treatment group consists of teams whose associated product lines adopted the digital tracking system; the control group comprises teams not yet treated. The outcomes include total output, output per worker, numbers of workers and managers, the worker-to-manager ratio, and the number of new workers. We present only the core outcomes here; results for the remaining outcomes are shown in Figure A5. The sample is a team-bimonthly panel covering 38 teams from January 2020 to September 2024. The vertical dashed line marks the wave-specific timing of treatment.



(a) IHS(Total Output)



(b) IHS(Output per Worker)



(c) High Performers Transferred Out

Figure 3: Team-Level Heterogeneous Event Studies

Notes. This figure presents grouped event-study estimates and 95% confidence intervals from equation (4), estimated separately for teams with high-performing managers and other teams. Each panel corresponds to a distinct team-level outcome: total output, output per worker, and the number of high-performing workers transferred out. Red markers denote teams with high-performing managers, and gray markers denote other teams. Treatment is defined as exposure of a team's product line to the digital tracking system. The vertical dashed line indicates treatment timing. Control teams are those not yet treated. We present only the core outcomes here; results for the remaining outcomes are shown in Figure A7. The sample is a team-bimonthly panel of 38 teams from January 2020 to September 2024.

ONLINE APPENDIX

Full Theoretical Model

This appendix provides the full model underlying the theoretical framework in the main text. The model formalizes how digital tracking changes the information environment in which managers deploy human capital across tasks, and why the same information can generate different productivity and retention outcomes depending on managerial task-allocation capability.

Model Setup

Consider a production team that operates in a dynamic environment. The team is led by a fixed manager who, in each period $t = 0, 1, \ldots$, allocates a newly arriving unit mass of tasks across a unit mass of workers. Each worker $i \in [0, 1]$ can perform up to $k > 1$ units of tasks per period, so aggregate worker capacity is sufficient to complete all tasks. Tasks are of two types. A share $\lambda \in (0, 1)$ are *complex tasks*, denoted by C, which require non-routine judgment, technical reasoning, or careful execution. The remaining share $1 - \lambda$ are *menial tasks*, denoted by M, which are routine, standardized, and require limited discretion.

Worker–Task Match. The value created by a worker–task match depends on both the worker's skill and the type of task performed. A worker is either high-skill, denoted by H, or low-skill, denoted by L. The team is initially drawn from a general labor pool in which a share $\mu \in (0, 1)$ of workers are high-skill. As described below, the team's high-skill share may subsequently evolve as workers leave the team and are replaced. We assume $k\mu < 1$, so high-skill workers alone cannot complete the full task mass. We normalize the output from assigning a low-skill worker to any task to one. Relative to this benchmark, assigning a high-skill worker generates an additional surplus Δ_C on a complex task and Δ_M on a menial task, where $\Delta_C > \Delta_M > 0$. Thus, high-skill workers have an absolute advantage on both types of tasks, but their productivity advantage is larger on complex tasks.

Timeline. Within each period, the game unfolds as follows. First, tasks are assigned and current-period output is realized. The manager chooses a worker-level assignment $\{c_t(i), m_t(i)\}_{i\in[0,1]}$, where $c_t(i) \geq 0$ and $m_t(i) \geq 0$ denote the masses of complex and menial tasks assigned to worker i, respectively. An assignment is feasible if it clears the task mass and respects individual capacity:

$$\int_0^1 c_t(i)\, di = \lambda, \quad \int_0^1 m_t(i)\, di = 1 - \lambda, \quad \text{and} \quad c_t(i) + m_t(i) \leq k \quad \forall i \in [0,1]. \tag{7}$$

Let c_t^H and m_t^H denote the average masses of complex and menial tasks assigned to high-skill workers, as induced by a feasible assignment. The team's output in period t can then be written as

$$y_t = 1 + \eta_t(c_t^H \Delta_C + m_t^H \Delta_M), \tag{8}$$

where η_t denotes the share of high-skill workers at the beginning of period t.

Second, after output is realized, high-skill workers may leave the team depending on the workload assigned to them. Let $w_t^H \equiv c_t^H + m_t^H$ denote the average workload of high-skill workers induced by the assignment. The fraction of high-skill workers who leave for workload-related reasons is $d_t = \tau \cdot \max\{w_t^H - \bar{w}, 0\}$, where $\bar{w}$ denotes the sustainable workload level and $0 < \tau < 1/(k - \bar{w})$, so that $d_t < 1$ for any feasible assignment. This formulation captures the idea that excessive workload makes high-skill workers more likely to transfer out of the team. To focus on the empirically relevant case in which managers can protect high-skill workers from overload but can also overuse them, we assume

$$\max\{\lambda/\mu, (1 - k(1-\mu))/\mu\} < \bar{w} < 1.$$

The lower-bound restriction implies that the manager can keep high-skill workers at a sustainable workload while assigning them all complex tasks and leaving a feasible residual task

mass for low-skill workers. The upper-bound restriction implies that using high-skill workers up to their full capacity necessarily induces such departures. For simplicity, we also assume that low-skill workers do not leave for workload-related reasons.

Finally, after workload-related departures occur, an exogenous fraction $\rho \in (0, 1)$ of the remaining workforce turns over. All departing workers are replaced by workers drawn from the general labor pool, where the share of high-skill workers is μ. It follows that the team's high-skill share evolves as

$$\eta_{t+1} = (1 - \rho)\eta_t + \rho\mu - (1 - \rho)(1 - \mu)d_t\eta_t. \tag{9}$$

This law of motion makes explicit the dynamic cost of overloading high-skill workers. Workload-related departures lower the team's future high-skill share because departing high-skill workers are replaced from the general labor pool.

Manager Type. Managers differ in whether they internalize the dynamic retention cost created by overloading high-skill workers. There are two manager types. Both types understand the productivity differences between high- and low-skill workers across tasks; they differ only in how they evaluate the effect of task assignments on future team composition. A *myopic manager* only cares about current-period output. Under the prevailing information structure, she chooses a feasible worker-level assignment $\{c_t(i), m_t(i)\}_{i\in[0,1]}$ to maximize expected current-period output as specified in (8), treating the current team composition η_t as fixed and ignoring the effect of the induced workload w_t^H on future retention. A *good manager*, in contrast, recognizes that overloading high-skill workers can reduce the future stock of high-skill labor. To capture this long-run orientation while keeping the model tractable, we assume that she chooses a stationary assignment policy to maximize the team's steady-state output. Let c^H, m^H, $w^H = c^H + m^H$, $d = \tau \max\{w^H - \bar{w}, 0\}$, and η denote the corresponding

steady-state values. In steady state, $\eta_{t+1} = \eta_t = \eta$, so the law of motion in (9) implies

$$\eta = \frac{\rho\mu}{\rho + (1-\rho)(1-\mu)d}. \tag{10}$$

Thus, under the information structure in place, a good manager chooses a stationary worker-level assignment policy to maximize expected steady-state output as given by

$$y = 1 + \eta(c^H \Delta_C + m^H \Delta_M), \tag{11}$$

subject to feasibility and the steady-state composition condition in (10).

Implications of Digital Tracking

We now characterize task assignments and team performance before and after the introduction of the digital tracking system. The system changes the manager's information environment by revealing worker-specific performance information that was previously unavailable. We model this change as making worker types observable: before tracking, managers cannot identify individual workers' types; after tracking, they can distinguish high- and low-skill workers and tailor task assignments accordingly.

The Pre-Tracking Benchmark. Consider first the pre-tracking setting. Although managers know that workers differ in productivity, they lack the individual-level information needed to act on those differences. Hence, task assignments cannot systematically vary with worker type. In particular, high-skill workers receive the same average task mix and workload as the team as a whole: $c_t^H = \lambda$, $m_t^H = 1 - \lambda$, and $w_t^H = 1$. This assignment policy implies a workload-related departure rate $d^0 = \tau(1 - \bar{w})$. Using the steady-state composition equation in (10), the corresponding high-skill share is

$$\eta^0 = \frac{\rho\mu}{\rho + (1-\rho)(1-\mu)\tau(1-\bar{w})}. \tag{12}$$

Thus, the steady-state per-period output before tracking is

$$y^0 = 1 + \eta^0 \left[\lambda \Delta_C + (1-\lambda)\Delta_M\right]. \tag{13}$$

The pre-tracking benchmarks in (12) and (13) are common to both manager types because, absent information about individual workers' abilities, good managers cannot exploit their superior task-allocation skill to improve assignment decisions.[16]

Post-Tracking Team Performance. After tracking is introduced, worker types become observable. Consider first the allocation chosen by a good manager. Since high-skill workers generate a larger surplus on complex tasks than on menial tasks, the manager assigns high-skill capacity to complex tasks first. Under the maintained assumption $\bar{w} > \lambda/\mu$, the candidate stationary policy that preserves the benchmark high-skill share can assign all complex tasks to high-skill workers without exceeding the sustainable workload level. Because $\Delta_M > 0$, the manager uses any remaining sustainable high-skill capacity on menial tasks. When Δ_M is sufficiently small, however, she does not find it worthwhile to increase current output by pushing high-skill workers above $\bar{w}$, because doing so would reduce the team's future high-skill share. Her optimal stationary policy therefore keeps high-skill workload at the sustainable level, with task mix $c_G^H = \lambda/\mu$ and $m_G^H = \bar{w} - \lambda/\mu$. This policy induces no workload-related departures, so the steady-state high-skill share and per-period output under a good manager are

$$\eta_G = \mu, \qquad y_G = 1 + \lambda\Delta_C + (\mu\bar{w} - \lambda)\Delta_M. \tag{14}$$

Proposition 1. *Suppose Δ_M is sufficiently small. Under the optimal stationary policy of a good manager after tracking, $\eta_G > \eta^0$ and $y_G > y^0$.*

[16] The model can accommodate pre-tracking productivity differences across teams by allowing managers to observe imperfect signals about workers' skill types before tracking. We abstract from this extension to keep the comparison with the post-tracking information environment sharp.

PROOF. From (12) and $\bar{w} < 1$, the uniform pre-tracking allocation generates workload-related departures among high-skill workers, so $\eta^0 < \mu = \eta_G$. Intuitively, tracking gives the good manager the information needed to protect high-skill workers from excessive workload. As for output, subtracting (13) from (14) yields

$$y_G - y^0 = \lambda\Delta_C(1-\eta^0) + \Delta_M\left[\mu\bar{w} - \lambda - \eta^0(1-\lambda)\right].$$

The first term is strictly positive because tracking allows the good manager to assign the full mass of complex tasks to high-skill workers. The second term captures the change in menial tasks assigned to high-skill workers. Hence, when Δ_M is sufficiently small–that is, when skill differences matter sufficiently little for menial tasks–the complex-task gain dominates, and steady-state output rises relative to the pre-tracking benchmark. □

We next consider the allocation chosen by a myopic manager. Once worker types are observable, a myopic manager also assigns high-skill workers to complex tasks first because $\Delta_C > \Delta_M$. The difference lies in how she uses any remaining high-skill capacity. Since she maximizes current-period output and $\Delta_M > 0$, she assigns high-skill workers as many additional menial tasks as feasibility allows. Under the maintained assumption $k\mu < 1$, this policy uses high-skill workers up to their individual capacity, so it induces $w_t^H = k$. The steady-state high-skill share under a myopic manager is therefore

$$\eta_B = \frac{\rho\mu}{\rho + (1-\rho)(1-\mu)\tau(k-\bar{w})}. \tag{15}$$

The corresponding steady-state per-period output is

$$y_B = 1 + \Delta_C \min\{\lambda, k\eta_B\} + \Delta_M \max\{k\eta_B - \lambda, 0\}. \tag{16}$$

Proposition 2. *Under the post-tracking policy of a myopic manager, $\eta_B < \eta^0$. Moreover, if $\bar{w}$ is sufficiently close to one and ρ is sufficiently small, then $y_B < y^0$.*

PROOF. Because high-skill workload is higher under the myopic post-tracking policy than under the pre-tracking allocation ($k > 1$), (15) has a larger denominator than (12), so $\eta_B < \eta^0$. As for output, subtracting (13) from (16) yields

$$y_B - y^0 = \Delta_C \left[\min\{\lambda, k\eta_B\} - \lambda\eta^0\right] + \Delta_M \left[\max\{k\eta_B - \lambda, 0\} - (1-\lambda)\eta^0\right].$$

The first term captures the change in complex tasks assigned to high-skill workers, while the second term captures the change in menial tasks. The condition $k\eta_B < \lambda\eta^0$ is sufficient for both bracketed terms to be negative. Using (12) and (15), this condition is equivalent to

$$\rho(k-\lambda) < (1-\rho)(1-\mu)\tau\left[\lambda(k-\bar{w}) - k(1-\bar{w})\right].$$

It is easy to see that for $\bar{w}$ sufficiently close to one and ρ sufficiently small, the inequality holds. Intuitively, as $\bar{w} \to 1$, the pre-tracking uniform allocation generates only mild workload-related departures, while the myopic post-tracking policy still pushes high-skill workers to capacity $k > 1$. A small ρ then makes these additional talent losses persistent, because lost high-skill workers are replenished only slowly through draws from the general labor pool. Together, these forces generate a sufficiently large decline in the steady-state high-skill share that the productivity loss from weakened team composition outweighs the direct allocation gain from tracking. □

Empirical Predictions

The model yields four empirical implications. First, tracking should increase scale more reliably than average productivity. Although the baseline model holds the mass of tasks and workers fixed and therefore does not formally derive a scale effect, such an effect follows immediately if tracking expands the amount of work a manager can supervise. At the same time, the model shows that average productivity depends on how managers use the new information. In the data, this implies that total output and span of control should increase

after adoption, while output per worker need not increase on average.

Second, tracking should have heterogeneous effects on productivity across teams. When managers use the new information effectively, tracking improves worker–task matching and raises team output. When managers instead use the same information to over-rely on high-performing workers, tracking can weaken future team composition and reduce output. Empirically, we therefore expect output per worker to increase more in teams led by high-performing managers, with weaker gains or even declines in other teams.

Third, these productivity effects should be accompanied by changes in worker retention and team composition. The model predicts that high-performing workers are better retained when managers use tracking to keep their workload sustainable. By contrast, if tracking leads managers to concentrate too much work on these workers, their likelihood of leaving should increase. In the data, this implies that teams led by high-performing managers should better preserve their high-performing workers after tracking is introduced. Because the empirical setting features team expansion and new hiring, this prediction is best interpreted in relative terms: the high-performing-worker share need not mechanically rise in these teams, but it should deteriorate less than in other teams.

Fourth, the task-allocation mechanism should be visible in post-tracking task records and worker-experience measures. The theory does not imply that high-performing workers should simply receive more work under good managers. Instead, good managers use tracking to allocate workload more selectively, especially by limiting excessive or low-value assignments to high-performing workers. Empirically, we therefore expect high-performing workers in high-performing-manager teams to face lower total workload and fewer menial-task assignments, without necessarily receiving more complex-task volume. To the extent that this allocation reduces overload, we also expect these workers to exhibit fewer downstream consequences of excessive workload.

Further Details on the Firm's Wage Policy

Determination of the Remuneration Pool

The firm's remuneration pool determination process evolved through three distinct phases during our sample period.

Phase I: 2020–2022. During this period, the remuneration pool for each product line was calculated by multiplying the line's monthly output value by a pre-determined contribution factor intended to capture the role of labor in the production process. The output value refers to the monetary valuation of projects completed by the product line each month, as estimated by the firm's top management.

To illustrate, suppose a product line fulfills two client orders in a given month, with per-unit market values of 2,000 CNY and 4,000 CNY, respectively. If top management assesses that the product line's components account for 50% of the market value in both cases, and 100 units of each are completed, then the total output value is: $2,000 \times 0.5 \times 100 + 4,000 \times 0.5 \times 100 = 2\,\text{million}$ CNY. The remuneration pool is then calculated as a fixed fraction of this total output value—referred to as the *contribution ratio*, which is set by the HR department to reflect workers' contribution to the production process of the product line. For example, with a contribution ratio of 2%—a typical value during our sample period—the monthly remuneration pool for the product line would be: $2\,\text{million CNY} \times 0.02 = 40,000\,\text{CNY}$. Generally, contribution ratios can vary across departments but usually remain the same for all product lines within the same department. These ratios are updated annually; each January, the HR department conducts a revision based on feedback from the engineering team overseeing all production activities. This feedback helps to identify changes in both absolute and relative efficiency in the production processes related to different departments and product lines, with the goal of maintaining stable average compensation levels over time.

Phase II: 2023. A key shortcoming of the prior determination process of the remuneration pool was the high variability it introduced into workers' compensation (particularly their effective hourly wages) due to fluctuations in the market value of orders. For instance, two projects requiring similar effort and time may carry vastly different market values depending on demand conditions or whether the client is a government or private entity.

To reduce this variability, two departments—Medium Voltage Gas Insulated Switchgear and Medium Voltage Air Insulated Switchgear—piloted a piece-rate remuneration system in 2023. Under this approach, the remuneration pool was based on the number of completed units or tasks (e.g., wire connections), rather than the estimated market value of orders. Other departments continued using the value-based method.

Phase III: 2024. Beginning in 2024, the piece-rate system was rolled out across all departments, standardizing wage determination based on completed output volume.

Determination of Individual Wages

After the remuneration pool of a product line is determined by the senior leadership of the firm, it is allocated to teams based on their relative contribution. Within each team, the team manager is responsible for distributing the funds to individual workers. This two-tier allocation process has remained stable over time.

The team-level distribution can be further broken down into two steps. First, each worker receives a fixed wage based on their certified skill tier, as recorded in the firm's system. Second, the remaining funds are distributed as performance-based bonuses. For example, suppose a team has 20,000 CNY to allocate across five workers. If two senior workers each receive a fixed wage of 3,000 CNY and three junior workers receive 1,500 CNY each, the total fixed wage component is: $(2 \times 3,000) + (3 \times 1,500) = 10,500$ CNY. The remaining 9,500 CNY is allocated as discretionary performance pay. This distribution is based primarily on the team manager's feedback and, to a lesser extent, the product line manager's own subjective assessment of individual performance.

Details of Balance Tests

First, we compare pre-treatment characteristics between ever-treated and never-treated teams using the following regression:

$$\text{Characteristic}_{ht} = \alpha + \beta \cdot \text{EverTreated}_h + \epsilon_{ht}, \tag{17}$$

where $\text{Characteristic}_{ht}$ denotes a team-level variable measured in bimonthly period t (prior to the implementation period), and EverTreated_h is an indicator for whether team h was ever exposed to the tracking system. We find no significant differences for most observables, though ever-treated teams have a slightly higher share of younger, less-experienced workers. To account for this, all main regressions include controls for time-varying team composition: average worker and manager age and experience, and the shares of female, college-educated, and Han-ethnicity workers.

Second, we test for differential pre-treatment trends in team composition using the following model:

$$\Delta\text{Characteristic}_{ht} = \alpha + \beta \cdot \text{EverTreated}_h + \epsilon_{ht}, \tag{18}$$

where $\Delta\text{Characteristic}_{ht}$ denotes the first difference in a team characteristic for team h in period t. This approach allows us to detect whether treated teams were already on different trajectories prior to adoption. Results show no systematic pre-trend differences.

Third, we examine whether rollout order among treated teams can be predicted from pre-treatment changes in team characteristics. Let $W_h \in \{1, \ldots, 7\}$ denote the wave in which team h was first exposed (the rollout occurred in seven waves). We estimate an ordered logit model of the following form:

$$\mathbb{P}(W_h \leq w) = \frac{\exp(\theta_w - \Delta X'_{ht}\gamma)}{1 + \exp(\theta_w - \Delta X'_{ht}\gamma)}, \tag{19}$$

where ΔX_{ht} includes the first differences of team characteristics measured before treatment,

and θ_w represents wave-specific thresholds. We find no significant predictors of rollout order, suggesting that observable pre-treatment dynamics do not explain the timing of adoption.

Details of the Composition-Standardization Exercise

The main text uses a counterfactual outcome to ask how team performance would have evolved if each team's worker mix had remained fixed at its pre-implementation composition. We formalize this counterfactual as a composition-standardized performance measure. The measure holds workforce composition fixed while allowing average performance within worker groups to evolve over time.

Construction of the Composition-Standardized Measure

To construct the composition-standardized measure, we use performance pay as a proxy for worker contribution. Although performance pay is not identical to physical output, team-level productivity gains are typically reflected in a larger compensation pool. Within teams, workers with higher marginal contributions also tend to receive higher payments on average. Managerial discretion in task assignment or scoring may introduce noise, but such noise does not mechanically imply the treatment patterns we examine. We therefore use performance pay not to recover exact productivity magnitudes, but to assess the direction of performance changes and the extent to which treatment heterogeneity remains once workforce composition is held fixed.

Specifically, we construct an outcome variable Y_{jt}^{std} in four steps:

Step 1: Define worker groups. We partition the workforce into three mutually exclusive types (indexed by k) based on pre-tracking records and team tenure:

- **Old high performers (OH):** Incumbents present in the team prior to implementation who ranked in the top 50% of the pre-tracking performance distribution.
- **Old lower performers (OL):** Incumbents present prior to implementation who

ranked in the bottom 50% of the pre-tracking performance distribution.

- **New workers (N):** All entrants to the team after implementation, including both internal transfers and new hires.[17]

Step 2: Compute group-specific mean performance. For each team j and bimonthly period t, we compute the average performance pay, μ_{jtk}, for each worker group $k \in \{OH, OL, N\}$. Intuitively, this average captures within-Group performance in team j at time t.

Step 3: Compute baseline composition weights. We compute team-specific baseline weights, w_{jk}^{pre}, as the share of worker-period observations belonging to each group k in team j during the pre-implementation period ($t < t_j^*$, where t_j^* is team j's implementation period). These weights summarize each team's pre-implementation worker mix.

Step 4: Construct the standardized outcome. The final composition-standardized outcome is defined as the weighted average:

$$Y_{jt}^{std} = \sum_{k \in \{OH, OL, New\}} w_{jk}^{pre} \cdot \mu_{jtk}.$$

By construction, Y_{jt}^{std} fixes each team's workforce composition at its pre-implementation benchmark. Variation in Y_{jt}^{std} over time is therefore driven by shifts in within-type average performance (μ_{jtk}), rather than by changes in the relative shares of worker types.

Our standardization approach has two advantages over natural alternatives. First, specifications that flexibly control for observed team composition yield estimates similar to the baseline results, but remain difficult to interpret with respect to the role of retention. Second, restricting the sample to team-periods composed entirely of pre-implementation incumbents removes worker reallocation mechanically, but also leaves little evidence of treatment effects, making it difficult to relate those estimates to the productivity gains observed in teams led

[17] The new-worker category is defined independently of pre-tracking performance and therefore does not require classifying entrants by ability.

by high-performing managers. By adjusting for workforce composition directly in the outcome measure, the composition-standardization exercise isolates the compositional margin more transparently while retaining the full sample.

Results and Discussion

Table A20 reports estimates using Y_{jt}^{std} as the dependent variable. In Columns (1) and (3), the coefficient on *Implemented* remains positive and statistically significant, indicating a robust average effect of digital tracking on performance even after workforce composition is held fixed. In Columns (2) and (4), however, the interaction term *Implemented* $\times$ *Good Mgr.* becomes smaller and statistically insignificant. Thus, once the compositional margin is held fixed, the differential treatment effect by managerial type becomes much less pronounced.

These patterns suggest that differential retention is an important channel through which the productivity effects of digital tracking are realized over time. As documented in the Mechanisms section, teams led by high-performing managers are better able to retain a high-performing workforce after adoption, and this retention difference contributes to the divergence in realized performance across teams. At the same time, the composition-standardization exercise should not be interpreted as a sharp decomposition of the mechanisms at work. As formalized in the Theoretical Framework section, task allocation and worker retention can be endogenously linked: poor allocation can trigger worker dissatisfaction and subsequent transfers. Because the standardized outcome is constructed using the realized performance of workers who remain in the team, it may mute the acute productivity losses that initially induced those exits. For this reason, while the narrowing of the productivity gap in Table A20 highlights the importance of the compositional margin, it remains fully consistent with managerial task-allocation quality being the fundamental first mover that generates those compositional changes.

Details on Alternative Explanations

This appendix provides additional empirical evidence ruling out several alternative mechanisms that could potentially account for the heterogeneous retention and productivity patterns documented in the Mechanisms section. In particular, we examine whether these patterns can be explained by differences in compensation and promotion opportunities, differential engagement with the tracking system, variation in material availability, or other workplace conditions unrelated to task allocation.

Compensation and Promotion Opportunities

We first examine whether the improved retention of high-performing workers under high-performing managers can be explained by differential financial rewards or promotion prospects following the introduction of the tracking system. If managers responded to enhanced performance visibility by rewarding high-performing workers more generously, either through higher performance pay or more frequent promotions, this could account for the observed retention differences.

To assess the alternative channel of differential financial rewards, we estimate a DiD specification using worker-month panel data from January 2020 to September 2024. The outcome variable is the inverse hyperbolic sine of performance-based wages. The key coefficient of interest is on the interaction between indicators for exposure to the tracking system and supervision by a high-performing manager. Estimation is conducted separately for high-performing workers and for other workers. Columns (1) and (2) of Table A23 show that the interaction term is positive but statistically insignificant for high-performing workers ($p = 0.321$), indicating no differential increase in performance-based pay under high-performing managers. By contrast, the interaction effect is positive and statistically significant for other workers, corresponding to a 23% larger increase in performance pay in teams led by high-performing managers. This pattern suggests that differences in financial incentives do not explain the improved retention of high-performing workers.

We next assess whether the observed retention effects could instead be driven by differences in promotion opportunities. In our setting, workers may be promoted to team manager roles or transferred to non-production departments such as sales or logistics. Although frontline performance is likely an important consideration, these promotions also depend on a variety of factors, including the worker's social skills, personal networks, and perceived managerial potential. Using a similar empirical strategy, we replace the dependent variable with a monthly promotion dummy. Promotions are rare in our setting: only 42 occurred between January 2020 and September 2024, corresponding to a mean of 0.0002 at the worker-month level. As shown in columns (3)–(4) of Table A23, the interaction terms are statistically insignificant for both high-performing workers ($p = 0.395$) and other workers ($p = 0.275$). The sign for high-performing workers is negative, further reinforcing the lack of support for the alternative mechanism.

Engagement with the Tracking System

A second potential explanation is that high-performing managers may simply engage more effectively with the tracking system itself, independent of how tasks are allocated. As the HR personnel of our partner firm noted,"*Some managers see the tracking system as helpful for reducing management pressure, while others see it as just another top-down mandate.*" If managers differ in how actively they encourage workers to use the system or monitor compliance, improved system engagement could enhance performance and coordination, potentially accounting for the observed differences across teams.

To examine this possibility, we exploit a feature of the tracking system under which tasks are automatically terminated at midnight if workers fail to log completion manually. We interpret the absence of such auto-terminated tasks as an indicator of accurate system usage, which may reflect stronger managerial oversight or encouragement. Using this information, we construct a daily worker-level indicator equal to one if all tasks assigned to the worker were properly closed on that day, and zero otherwise. We then estimate equation (1) using

this indicator as the outcome variable.

Column (1) of Table A24 shows that workers supervised by high-performing managers are 5.1 percentage points more likely to have all tasks properly recorded on a given day. Disaggregating by worker type, Columns (2) and (3) show that this effect is concentrated among high-performing workers, for whom the estimated increase is 5.9 percentage points and statistically significant. In contrast, no corresponding effect is observed for workers outside the high-performing group.

While these results indicate that high-performing managers are modestly more successful in promoting accurate use of the tracking system, particularly among high-performing workers, this channel alone cannot account for the main empirical patterns. If improved system engagement were the primary driver of productivity gains, such gains would be expected to be concentrated among the same group of workers. However, estimating equation 6 separately by worker type shows that productivity gains under high-performing managers are observed across worker groups and are, if anything, more pronounced among workers outside the high-performing group (Table A25). This evidence suggests that differential engagement with the tracking system is unlikely to be the main mechanism behind the heterogeneous productivity effects.

Material Availability

We next consider whether differences in material availability could explain heterogeneous productivity effects. As discussed in the Empirical Context section, team managers are responsible for monitoring material needs and coordinating procurement when shortages arise. If high-performing managers were more effective at preventing shortages, this could lead to superior outcomes after tracking is introduced.

To assess this channel, we estimate equation (1) using a binary indicator for whether a worker experienced a material shortage on a given day. Columns (5)–(8) of Table A24 show no statistically significant differences in the incidence of shortages between workers

under high-performing versus other managers. A likely explanation for this null result is that material shortages are primarily driven by upstream or firm-wide constraints, such as supplier reliability or centralized inventory policies, over which individual team managers have limited influence. As a result, variation in material availability does not appear to be a major channel through which managerial ability shapes the effects of the tracking system.

Other Workplace Conditions

Lastly, we examine whether other workplace conditions unrelated to task allocation could account for the differential retention of high-performing workers under high-performing managers. To assess this possibility, we use survey data collected in September 2024, which capture a broad set of worker perceptions that are plausibly relevant for retention but not directly tied to task allocation. These include satisfaction with promotion prospects and benefits, perceptions of managerial communication and cooperation, and beliefs about the importance of one's job to the firm or to society.

We estimate a series of cross-sectional regressions using each survey measure as the dependent variable. The key explanatory variables are an indicator for high-performing workers, an indicator for being supervised by a high-performing manager at the time of the survey, and their interaction. The interaction term captures whether high-performing managers differentially affect these broader workplace perceptions among high-performing workers. Across all outcomes, we find no statistically significant interaction effects (Table A26). This evidence suggests that differences in these broader workplace conditions are unlikely to explain the heterogeneous retention patterns documented in the main analysis.

Additional Tables and Figures

Table A1: Summary Statistics for Administrative Data (Team × Bimonthly Period level)

Variable	Obs. #	Mean	Std. Dev.	10%	25%	50%	75%	90%
Manager %Female	1029	0.16	0.35	0	0	0	0	1
Worker %Female	1029	0.18	0.27	0	0	0	0.26	0.67
Manager %Ethnic Han	1029	0.95	0.20	1	1	1	1	1
Worker %Ethnic Han	1029	0.98	0.05	0.92	1	1	1	1
Manager %CCP/CYL Member	1029	0.40	0.42	0	0	0.33	1	1
Worker %CCP/CYL Member	1029	0.26	0.16	0.07	0.15	0.24	0.33	0.46
Manager %College	1029	0.20	0.34	0	0	0	0.33	1
Worker %College	1029	0.21	0.16	0	0.08	0.19	0.30	0.43
Average Manager Age	1029	35.23	7.19	25.20	30.33	34.75	40.00	45.00
Average Worker Age	1029	33.16	5.83	27.04	29.07	31.94	35.55	42.12
Average Manager Experience	1029	8.39	6.36	1.50	3	7.50	12	16
Average Worker Experience	1029	4.23	4.38	0.66	1.46	2.65	4.84	11.17

Notes. This table summarizes demographic characteristics used as control variables in the team-level bimonthly panel dataset. The share of female managers (workers) refers to the proportion of managers (workers) in each team and period who are female. The share of ethnic Han workers denotes the proportion of team members belonging to the Han ethnic group. The share of CCP/CYL members measures the proportion of workers who are members of the Chinese Communist Party or the Chinese Communist Youth League. Manager age and worker age capture the average age of managers and non-managerial workers, respectively, within each team and period. Manager experience and worker experience refer to the average years of work experience for managers and workers, respectively.

Table A2: Summary Statistics for Production Record Data (Worker-Day Level)

Variable	Obs. #	Mean	Standard Deviation	10%	25%	50%	75%	90%
Panel A. Task Allocation								
Total Workload	19656	1024.12	2295.05	0.36	66.00	298.58	879.96	2658.19
Complex Task Workload	19656	393.93	1687.05	0.00	0.00	0.00	137.46	799.84
Menial Task Workload	19656	630.19	1593.66	0.00	0.00	119.75	474.76	1679.94
Total Task Count	19656	15.21	27.95	1	2	5	16	39
Complex Task Count	19656	3.65	10.87	0	0	0	3	10
Menial Task Count	19656	12.57	27.01	0	0	3	13	33
Panel B. Production Outcomes								
Quality Defects	19656	0.34	0.47	0	0	0	1	1
Non-Compliance	19656	0.74	0.44	0	0	1	1	1
No Material Shortage	19656	0.49	0.50	0	0	0	1	1
Panel C. Worker and Manager Quality								
High-Performing-Manager Supervision	19656	0.09	0.29	0	0	0	0	0
High Performers	19656	0.55	0.50	0	0	1	1	1
Low Performers	19656	0.19	0.39	0	0	0	0	1

Notes. This table summarizes the worker-day level panel dataset used for the mechanism analysis. Panel A reports measures of task allocation constructed from the production record data. Total workload refers to the sum of the expected durations of all tasks assigned to a worker on a given day. Complex and Menial Task Workload capture the expected durations of only complex or only menial tasks, respectively. Total task count is the number of tasks assigned, while Complex and Menial Task Count refer to the number of complex or menial tasks assigned, respectively. Panel B presents production outcome variables, also constructed from the production record data. Quality Defects is a binary indicator for whether any task assigned to the worker was later reassigned. Non-Compliance is a binary indicator for whether the worker had at least one task auto-terminated at midnight. No Material shortage is a binary indicator for whether the worker experienced no material shortage on that day. Panel C includes measures of manager and worker quality used in the heterogeneity analysis.

Table A3: Correlation between Task Complexity and Mistake Likelihood

VARIABLES	(1)	(2)	(3)	(4)
	Any Mistake		Number of Mistakes	
	OLS	Probit	OLS	Poisson
Complex	0.042	0.285	0.030	0.314
	(0.001)	(0.008)	(0.002)	(0.013)
Observations	409,883	409,883	409,883	409,883
R-squared	0.004		0.001	
Sample Mean	0.065	0.065	0.087	0.087

Notes. Each observation corresponds to a task. The dependent variable in Columns (1) and (2) is an indicator for whether the task was ever performed incorrectly. The dependent variable in Columns (3) and (4) is the number of times the task was redone before it was completed. The independent variable is an indicator for whether the task is defined as complex.

Table A4: Summary Statistics for Survey Outcomes

Variable	Obs. #	Mean	Standard Deviation	10%	25%	50%	75%	90%
Panel A. Worker Responses								
Dissatisfaction with Promotion Chance	727	2.50	0.98	1	2	3	3	4
Dissatisfaction with Benefits Provided	727	2.39	1.04	1	1	3	3	4
Dissatisfaction with Workload	727	2.55	0.95	1	2	3	3	4
Mgr. Good Scheduling	727	3.09	0.83	2	3	3	4	4
Mgr. Good Cooperation	727	3.24	0.75	2	3	3	4	4
Important for Company	727	3.08	0.83	2	3	3	4	4
Important for Society	727	3.10	0.83	2	3	3	4	4
Comm. Smoothness with Mgr.	727	3.35	0.67	3	3	3	4	4
Comm. Times with Mgr.	727	10.81	5.50	3	7	10	15	20
Panel B. Manager Responses (Claims)								
Important for Company	110	3.12	0.80	2	3	3	4	4
Important for Society	110	3.11	0.78	2	3	3	4	4
Satis. with Work Status	110	3.30	0.78	2	3	3	4	4
Gratitude for the Company	110	3.54	0.69	3	3	4	4	4
Goal Unclear	110	1.71	0.97	1	1	1	2	3
Dissatisfaction with Promotion Chance	110	2.14	0.87	1	1	2	3	3
Dissatisfaction with Benefits Provided	110	2.11	1.02	1	1	2	3	4
Dissatisfaction with Workload	110	2.47	0.82	1	2	3	3	3
Workplace Conflict	110	1.43	0.77	1	1	1	2	3
Effort-reward Imbalance	110	1.97	0.92	1	1	2	3	3
Occupational Suitability	110	3.10	0.77	2	3	3	4	4
Panel C. Manager Responses (Beliefs)								
Experience Self Eval.	110	8.51	1.30	7	8	9	9	10
Skills Self Eval.	110	8.66	1.36	7	8	9	9	10
Expert Proficiency Self Eval.	110	8.48	1.55	7	8	9	10	10
Skill Versatility Self Eval.	110	8.55	1.37	7	8	9	9	10
Learning Ability Self Eval.	110	8.57	1.47	7	8	9	10	10
Teamwork Ability Self Eval.	110	8.83	1.41	7	8	9	10	10
Experience Importance	110	8.52	1.30	7	8	9	10	10
Skill Importance	110	8.78	1.18	7	8	9	10	10
Expert proficiency Importance	110	8.56	1.30	7	8	9	10	10
Skill Versatility Importance	110	8.51	1.49	7	8	9	10	10
Learning Ability Importance	110	8.95	1.12	7	8	9	10	10
Teamwork Ability Importance	110	9.30	1.06	8	9	10	10	10

Notes. This table summarizes the survey responses. Panel A presents worker responses. Each item, except for "Communication Time with Manager," is a self-reported agreement with a specific statement, recorded on a four-point Likert scale: 1 (strongly disagree), 2 (disagree), 3 (agree), and 4 (strongly agree). "Communication Time with Manager" is a continuous measure indicating the average number of minutes per day the worker reported spending in direct communication with their team manager. Panels B and C report manager responses. Panel B includes managers' self-assessments on the same four-point Likert scale. Panel C presents managers' self-evaluations of their own abilities and their perceived importance of various skills. These responses are measured on a ten-point scale ranging from 1 (not at all important) to 10 (extremely important).

Table A5: The Effects of High-performing Managers on Team Performance

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	IHS(Output per Worker)			Output per Worker		
	Pooled	Before	After	Pooled	Before	After
High-Performing Manager	0.120	0.139	0.111	1,490.638	1,143.668	1,665.356
	(0.066)	(0.033)	(0.080)	(947.362)	(297.851)	(1,214.953)
Observations	933	268	659	933	268	659
R-squared	0.690	0.793	0.715	0.683	0.831	0.689
Team Controls	Yes	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. This table reports estimates of the effects of being led by a high-performing manager on the team productivity. The high-performing manager indicator equals one if the team was led by a manager whose average pre-tracking leader allowance ranked in the top 25% of this distribution. The unit of observation is a team-bimonthly spell, and the sample spans January 2020 to September 2024. All regressions include team and period fixed effects, along with time-varying team-level covariates: the shares of workers who are female, ethnic Han, and college-educated, as well as the average age and tenure of both workers and managers. Standard errors are clustered at the team level.

Table A6: Correlation between Team Characteristics and Manager Type

VARIABLES	(1) Worker per Manager	(2) Team Size	(3) No. of Managers	(4) Frac. of High Performers	(5)
				Top 50%	Top 75%
High-Performing Managers	-0.622	3.283	0.543	0.024	-0.013
	(1.228)	(3.544)	(0.361)	(0.037)	(0.034)
Observations	38	38	38	38	38
R-squared	0.007	0.023	0.059	0.012	0.004
Sample Mean	7.71	14.90	1.79	0.69	0.65

Notes. Each observation is a team. The dependent variables are computed using only pre-implementation data and correspond to the team's average span of control (number of workers per manager), total team size, number of managers, and fraction of high-performers under alternative definitions. The key independent variable is an indicator equal to one if the team had at least one high-performing manager prior to system implementation.

Table A7: Correlation between Task Composition and Manager Type

VARIABLES	(1) No. of Complex Tasks	(2) No. of Tasks	(3) Complex Tasks Ratio
High-Performing Managers	32.080	-516.589	0.061
	(82.666)	(788.435)	(0.117)
Observations	27	27	27
R-squared	0.006	0.017	0.011
Sample Mean	125.73	1074.05	0.27

Notes. Each observation is a team. The dependent variables are constructed using only post-implementation task records, that is, periods in which the team is observed in the digital tracking system. For each team, we compute the average number of complex tasks, the average total number of tasks, and the average share of tasks classified as complex across observed team-bimonthly periods. The key independent variable is an indicator equal to one if the team had at least one high-performing manager prior to system implementation.

Table A8: Effect of Managerial Quality on Task Allocation Decisions (Actual Worktime)

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	IHS(Workload)				Workload			
	All Workers	High Performers	Low Performers	New Workers	All Workers	High Performers	Low Performers	New Workers
Supervised by a High-Performing Manager	-0.309	-0.519	0.475	0.914	-835.023	-1,174.301	252.524	-41.546
	(0.208)	(0.223)	(0.416)	(0.758)	(500.346)	(634.243)	(271.072)	(963.429)
Observations	12,962	8,169	2,748	1,920	12,962	8,169	2,748	1,920
R-squared	0.443	0.454	0.449	0.590	0.157	0.161	0.292	0.320
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. This table reports robustness checks for the effect of managerial quality on workload allocation, using realized rather than expected task durations. The dependent variable is the sum of the actual durations of a worker's assigned tasks on a given day. Columns (1)–(4) use the inverse hyperbolic sine (IHS) transformation of the dependent variable; Columns (5)–(8) use the raw value. All specifications are estimated using OLS. The unit of observation is a worker-day. Columns (1) and (5) include all workers; Columns (2) and (6) restrict to high performers; Columns (3) and (7) to low performers; and Columns (4) and (8) to new hires who joined after system implementation. All regressions include worker and date fixed effects. Standard errors are clustered at the worker level.

Table A9: Analysis Separating Complex and Menial Tasks

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	IHS(Complex Workload)				IHS(Menial Workload)			
	All Workers	High Performers	Low Performers	New Workers	All Workers	High Performers	Low Performers	New Workers
Supervised by a High-Performing Manager	-0.041	-0.123	0.310	2.148	-0.340	-0.501	0.254	2.225
	(0.148)	(0.197)	(0.220)	(0.687)	(0.211)	(0.241)	(0.289)	(1.339)
Observations	12,962	8,169	2,748	1,920	12,962	8,169	2,748	1,920
R-squared	0.740	0.753	0.707	0.818	0.673	0.667	0.681	0.806
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Variable	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
	Complex Task Number				Menial Task Number			
	All Workers	High Performers	Low Performers	New Workers	All Workers	High Performers	Low Performers	New Workers
Supervised by a High-Performing Manager	-0.484	-0.716	0.026	5.213	-6.657	-6.258	-6.570	-1.925
	(0.500)	(0.785)	(0.335)	(2.213)	(2.204)	(2.440)	(5.473)	(3.959)
Observations	12,962	8,169	2,748	1,920	12,962	8,169	2,748	1,920
R-squared	0.406	0.416	0.437	0.542	0.512	0.510	0.559	0.695
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. This table reports regressions of workers' daily workload and task counts for complex and simple tasks on an indicator for supervision by a high-performing manager. The unit of observation is a worker-day, with the sample covering January 2022 to July 2024. Some observations are dropped in grouped regressions due to becoming singletons when including date fixed effects. All regressions include worker and date fixed effects. Standard errors are clustered at the worker level.

Table A10: Balance Test for Task Composition

VARIABLES	(1)	(2)	(3)	(4)
	Implement Order			
	OLS	ologit	OLS	ologit
Task Complex Ratio	-0.386	-0.350		
	(0.364)	(0.387)		
Task Complex Ratio (first-diff)			-1.133	-1.048
			(0.899)	(0.928)
Observations	222	222	166	166
R-squared	0.005		0.010	

Notes. An observation is a team-bimonthly period. Only observations after the implementation (having records in the system) are included. The dependent variable is the implement order of the teams, with the earliest-implemented teams as 1 and the latest as 7. The independent variable for Column (1)-(2) is the level of task complex ratio (No. of complex tasks/No. of all tasks) and that for Column (3)-(4) is the first-difference of it.

Table A11: Balance Test

Variable	(1) Mean of Never Takers	(2) β of Ever Takers(Levels)	(3) β of Ever Takers(First-Diff)	(4) Timing of Implementation
Manager %Female	0.0495	0.2079	0.0006	4.2171
		(0.1856)	(0.0718)	(5.1605)
Worker %Female	0.0745	0.0532	-0.0024	6.3740
		(0.1322)	(0.0518)	(7.5534)
Manager %Ethnic Han	0.9904	-0.0022	0.0000	0.0000
		(0.0118)	(0.0000)	(0.0000)
Worker %Ethnic Han	0.9769	-0.0031	-0.0003	-20.6586
		(0.0120)	(0.0036)	(37.1164)
Manager %CCP/CLY Member	0.4692	-0.0969	-0.1707	-2.4894
		(0.1986)	(0.1417)	(2.0261)
Worker %CCP/CLY Member	0.1839	0.1487	-0.0214	-3.5355
		(0.1257)	(0.0498)	(4.7114)
Manager %College	0.1370	0.1815	0.0166	1.9138
		(0.1682)	(0.0669)	(3.7574)
Worker %College	0.1787	0.0574	-0.0037	11.3304
		(0.0930)	(0.0395)	(5.5094)
Manager %Age<35	0.5139	0.1733	-0.0850	5.1649
		(0.2235)	(0.1163)	(2.0846)
Manager %Age≥35	0.4861			
Worker %Age<30	0.2964	0.3239	-0.0590	1.4125
		(0.1423)	(0.0674)	(2.9796)
Worker %Age≥30	0.6476			
Manager %Experience<5	0.3811	0.1585	-0.0046	-4.1609
		(0.2062)	(0.1305)	(2.1071)
Manager %Experience≥5	0.6189			
Worker %Experience<5	0.6647	0.3097	0.0044	4.6497
		(0.1279)	(0.0340)	(4.8733)
Worker %Experience≥5	0.2582			

Notes. This table compares predetermined team characteristics between ever-treated and never-treated teams and examines differences by implementation timing. The sample covers the pre-implementation period from January 2020 to August 2020. Column (1) reports the mean of each characteristic for never-treated teams. Column (2) shows coefficients from regressions of each characteristic on an indicator for ever-treated teams (levels). Column (3) replicates Column (2) using first differences of the characteristics as dependent variables. Column (4) reports coefficients from an ordered logit regression of the order of system adoption on all characteristics. Standard errors are shown in parentheses.

Table A12: Average Treatment Effects on Key Outcomes (Pre-trend Subtracted)

VARIABLES	(1) IHS(Output)	(2) Workers	(3) IHS(Worker per Manager)
Implemented	0.694	4.828	0.048
	(0.173)	(1.519)	(0.078)
Observations	1,029	1,029	1,029
R-squared	0.794	0.749	0.611
Control Mean	12.28	14.30	2.79
Team Controls	Yes	Yes	Yes
Team FE	Yes	Yes	Yes
Period FE	Yes	Yes	Yes

Notes. This table reports estimates of the effect of the digital tracking system on team-level outcomes that exhibit pre-trends in the baseline event studies, using outcomes detrended by pre-implementation trends. Specifically, for each outcome we use the pre-implementation sample to estimate cohort-specific linear time trends, and then subtract the fitted trend from the outcome in all periods. We use this detrended outcome as the dependent variable in the regressions. The treatment indicator equals one if the product line associated with a team implemented the system during the period. The unit of observation is a team-bimonthly spell, and the sample covers January 2020 to September 2024. All columns are estimated using OLS. All regressions include team and group-specific period fixed effects, as well as time-varying team-level controls: shares of workers who are female, ethnic Han, and college-educated, and the average age and tenure of both workers and managers. Control Mean reports the average dependent variable value among untreated observations. Standard errors are clustered at the team level. Standard errors are clustered at team level.

Table A13: Average Treatment Effects on Team-Level Outcomes (All Outcomes)

Panel A. Output Variables

Variable	(1) IHS(Output)	(2) IHS(Output per Worker)
Implemented	0.326	0.042
	(0.125)	(0.036)
Observations	955	955
R-squared	0.810	0.679
Control Mean	12.36	9.83
Team Controls	Yes	Yes
Team FE	Yes	Yes
Period FE	Yes	Yes

Panel B. Human Resource Variables

Variable	(1) Team Size	(2) No. of Managers	(3) IHS(Worker per Manager)	(4) No. of New Workers	(5) High Performers Transferred In
Implemented	0.265	0.164	0.177	0.581	0.157
	(0.126)	(0.142)	(0.075)	(0.181)	(0.377)
Observations	955	955	951	910	692
R-squared			0.541		
Control Mean	17.01	2.03	2.73	1.52	0.48
Team Controls	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes

Variable	(6) High Performers Transferred Out	(7) Poor Performers Transferred In	(8) Poor Performers Transferred Out	(9) High Performers Turnover	(10) Poor Performers Turnover
Implemented	-0.380	0.157	0.034	0.002	0.002
	(0.327)	(0.519)	(0.538)	(0.003)	(0.005)
Observations	750	864	825	955	955
R-squared				0.151	0.255
Control Mean	0.42	0.17	0.16	0.00	0.01
Team Controls	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes

Notes. This table reports estimates of the average treatment effect of the digital tracking system on various team-level outcomes. The treatment indicator equals one if the product line associated with a team implemented the system during the period. The unit of observation is a team-bimonthly spell, and the sample covers January 2020 to September 2024. Panel A presents output-related variables, while Panel B focuses on human resource variables. Columns (1)–(8) in Panel B use Poisson regressions due to the count nature of the dependent variables; all other columns use OLS. In Poisson specifications, team-period observations with all-zero outcomes are excluded, resulting in smaller sample sizes. All regressions include team and period fixed effects and control for time-varying team-level covariates: the shares of workers who are female, ethnic Han, and college-educated, as well as the average age and tenure of both workers and managers. Control Mean reports the average dependent variable value among untreated observations. Standard errors are clustered at the team level.

Table A14: Heterogeneous Effects by Managerial Quality (All Outcomes)

Panel A. Output Variables

Variable	(1) IHS(Output)	(2) IHS(Output per Worker)
Implemented	0.367	-0.063
	(0.121)	(0.030)
Implemented × High-Performing Manager	-0.018	0.179
	(0.212)	(0.066)
Observations	955	955
R-squared	0.819	0.697
Control Mean	12.28	9.83
Team Controls	Yes	Yes
Team FE	Yes	Yes
Group × Period FE	Yes	Yes

Panel B. Personnel Variables

Variable	(1) Team Size	(2) No. of Managers	(3) IHS(Worker per Manager)	(4) No. of New Workers	(5) High Performers Transferred In
Implemented	0.424	0.208	0.279	0.738	0.696
	(0.088)	(0.100)	(0.130)	(0.295)	(0.792)
Implemented × High-Performing Manager	-0.138	-0.088	-0.064	-0.130	-0.557
	(0.149)	(0.189)	(0.164)	(0.369)	(0.842)
Observations	955	955	951	895	579
R-squared			0.573		
Control Mean	14.30	1.62	2.81	1.63	0.48
Team Controls	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes
Group × Period FE	Yes	Yes	Yes	Yes	Yes

Variable	(6) High Performers Transferred Out	(7) Low Performers Transferred In	(8) Low Performers Transferred Out	(9) High Performers Turnover	(10) Low Performers Turnover
Implemented	1.040	-0.056	-0.275	-0.002	-0.003
	(0.632)	(1.277)	(0.810)	(0.002)	(0.005)
Implemented * High-Performing Manager	-1.713	0.113	-0.120	0.008	0.006
	(0.816)	(1.398)	(1.039)	(0.006)	(0.009)
Observations	642	687	706	955	955
R-squared				0.179	0.286
Control Mean	0.42	0.17	0.16	0.01	0.02
Team Controls	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes
Group*Period FE	Yes	Yes	Yes	Yes	Yes

Notes. This table reports estimates of the effect of the digital tracking system and its interaction with managerial quality on team-level outcomes. Managerial quality is based on the average pre-tracking leader allowance. The high-performing manager indicator equals one if the manager ranked in the top 25% of this distribution. The treatment indicator equals one if the product line associated with a team implemented the system during the period. The managerial quality indicator equals one if the team had at least one high-performing manager at any point prior to being exposed to the system. The unit of observation is a team-bimonthly spell, and the sample covers January 2020 to September 2024. Panel A presents output variables, while Panel B reports human resource variables. Columns (1)–(8) in Panel B use Poisson regressions due to the count nature of the dependent variables; all other columns use OLS. In Poisson specifications, team-period observations with all-zero outcomes are excluded, resulting in smaller sample sizes. All regressions include team and group-specific period fixed effects, as well as time-varying team-level controls: shares of workers who are female, ethnic Han, and college-educated, and the average age and tenure of both workers and managers. Control Mean reports the average dependent variable value among untreated observations. Standard errors are clustered at the team level. Standard errors are clustered at team level.

Table A15: Heterogeneous Effects by Managerial Quality (Top 50% Criterion for Managers)

Panel A. Output Variables

Variable	(1) IHS(Output)	(2) IHS(Output per Worker)
Implemented	0.425	-0.059
	(0.150)	(0.034)
Implemented × High-Performing Manager	-0.149	0.156
	(0.207)	(0.065)
Observations	955	955
R-squared	0.823	0.690
Control Mean	12.05	9.82
Team Controls	Yes	Yes
Team FE	Yes	Yes
Group × Period FE	Yes	Yes

Panel B. Personnel Variables

Variable	(1) Team Size	(2) No. of Managers	(3) IHS(Worker per Manager)	(4) No. of New Workers	(5) High Performers Transferred In
Implemented	0.485	0.360	0.215	1.110	1.148
	(0.147)	(0.125)	(0.152)	(0.575)	(0.850)
Implemented × High-Performing Manager	-0.280	-0.262	-0.055	-0.728	-0.915
	(0.176)	(0.193)	(0.181)	(0.593)	(0.882)
Observations	955	955	951	910	572
R-squared			0.558		
Control Mean	11.79	1.42	2.69	1.55	0.42
Team Controls	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes
Group × Period FE	Yes	Yes	Yes	Yes	Yes

Variable	(6) High Performers Transferred Out	(7) Low Performers Transferred In	(8) Low Performers Transferred Out	(9) High Performers Turnover	(10) Low Performers Turnover
Implemented	1.352	-0.231	0.206	-0.003	-0.005
	(1.111)	(1.021)	(1.295)	(0.003)	(0.006)
Implemented × High-Performing manager	-2.232	0.387	-0.729	0.008	0.010
	(1.215)	(1.202)	(1.417)	(0.005)	(0.009)
Observations	653	693	698	955	955
R-squared				0.173	0.290
Control Mean	0.34	0.17	0.16	0.01	0.02
Team Controls	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes
Group × Period FE	Yes	Yes	Yes	Yes	Yes

Notes. This table reports estimates of the effect of the digital tracking system and its interaction with managerial quality on team-level outcomes. The high-performing manager indicator equals one if the average pre-tracking leader allowance of a manager ranked in the top 50% of this distribution. The treatment indicator equals one if the product line associated with a team implemented the system during the period. The managerial quality indicator equals one if the team had at least one high-performing manager at any point prior to being exposed to the system. The unit of observation is a team-bimonthly spell, and the sample spans January 2020 to September 2024. Panel A reports output-related outcomes; Panel B presents human resource outcomes. In Panel B, Columns (1)–(8) are estimated using Poisson regressions due to the count nature of the dependent variables; all other columns use OLS. In Poisson specifications, team-period observations with all-zero outcomes are excluded, resulting in smaller sample sizes. All regressions include team and group-specific period fixed effects, along with time-varying team-level covariates: the shares of workers who are female, ethnic Han, and college-educated, as well as the average age and tenure of both workers and managers. Control Mean reports the average value of the dependent variable among untreated observations. Standard errors are clustered at the team level.

Table A16: Heterogeneous Effects by Managerial Quality (Top 25% Criterion for Workers)

Panel A. Output Variables

Variable	(1) IHS(Output)	(2) IHS(Output per Worker)
Implemented	0.367	-0.063
	(0.121)	(0.030)
Implemented × High-Performing Manager	-0.018	0.179
	(0.212)	(0.066)
Observations	955	955
R-squared	0.819	0.697
Control Mean	12.28	9.83
Team Controls	Yes	Yes
Team FE	Yes	Yes
Group × Period FE	Yes	Yes

Panel B. Personnel Variables

Variable	(1) Team Size	(2) No. of Managers	(3) IHS(Worker per Manager)	(4) No. of New Workers	(5) High Performers Transferred In
Implemented	0.424	0.208	0.279	0.738	0.848
	(0.088)	(0.100)	(0.130)	(0.295)	(0.758)
Implemented × High-Performing Manager	-0.138	-0.088	-0.064	-0.130	-0.659
	(0.149)	(0.189)	(0.164)	(0.369)	(0.815)
Observations	955	955	955	895	575
R-squared			0.565		
Control Mean	14.30	1.62	2.79	1.63	0.49
Team Controls	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes
Group × Period FE	Yes	Yes	Yes	Yes	Yes

Variable	(6) High Performers Transferred Out	(7) Low Performers Transferred In	(8) Low Performers Transferred Out	(9) High Performers Turnover	(10) Low Performers Turnover
Implemented	0.902	-0.228	0.203	-0.003	-0.001
	(0.687)	(1.089)	(0.662)	(0.002)	(0.005)
Implemented × High-Performing Manager	-1.486	0.079	-0.517	0.007	0.008
	(0.814)	(1.230)	(1.003)	(0.004)	(0.010)
Observations	640	603	673	955	955
R-squared				0.179	0.302
Control Mean	0.44	0.17	0.14	0.01	0.02
Team Controls	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes
Group × Period FE	Yes	Yes	Yes	Yes	Yes

Notes. This table reports estimates of the effect of the digital tracking system and its interaction with managerial quality on team-level outcomes. High performers are defined as workers in the top 50% based on the share of pre-tracking months in which their performance pay ranked in the top 25% within their teams. The treatment indicator equals one if the product line associated with a team had implemented the system during the period. The managerial quality indicator equals one if the team had at least one high-performing manager at any point prior to being exposed to the system. The unit of observation is a team-bimonthly spell, and the sample covers the period from January 2020 to September 2024. Panel A presents output outcomes; Panel B presents human resource outcomes. Columns (1)–(8) in Panel B use Poisson regressions due to the count nature of the dependent variables; the remaining columns use OLS. Team-period observations with all-zero outcomes are dropped in Poisson regressions, leading to smaller sample sizes. All regressions include team and group-specific period fixed effects, along with time-varying team-level covariates: the shares of workers who are female, ethnic Han, and college-educated, as well as the average age and tenure of both workers and managers. Control Mean reports the average of the dependent variable among untreated observations. Standard errors are clustered at the team level.

Table A17: Heterogeneous Effects by Managerial Quality (No Switching Teams)

Panel A. Output Variables

Variable	(1) IHS(Output)	(2) IHS(Output per Worker)
Implemented	0.354	-0.062
	(0.123)	(0.028)
Implemented × High-Performing Manager	-0.127	0.195
	(0.221)	(0.088)
Observations	895	895
R-squared	0.808	0.691
Control Mean	12.28	9.83
Team Controls	Yes	Yes
Team FE	Yes	Yes
Group × Period FE	Yes	Yes

Panel B. Personnel Variables

Variable	(1) Team Size	(2) No. of Managers	(3) IHS(Worker per Manager)	(4) No. of New Workers	(5) High Performers Transferred In
Implemented	0.397	0.189	0.277	0.779	0.605
	(0.080)	(0.096)	(0.129)	(0.299)	(0.822)
Implemented × High-Performing Manager	-0.294	-0.258	-0.038	-0.049	-0.846
	(0.144)	(0.185)	(0.180)	(0.380)	(0.851)
Observations	895	895	891	840	552
R-squared			0.561		
Control Mean	14.30	1.62	2.81	1.63	0.49
Team Controls	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes
Group × Period FE	Yes	Yes	Yes	Yes	Yes

Variable	(6) High Performers Transferred Out	(7) Low Performers Transferred In	(8) Low Performers Transferred Out	(9) High Performers Turnover	(10) Low Performers Turnover
Implemented	1.286	0.060	-0.031	-0.002	-0.004
	(0.512)	(1.218)	(0.659)	(0.002)	(0.005)
Implemented × High-Performing Manager	-1.735	-0.149	-0.239	0.009	0.009
	(0.560)	(1.326)	(0.888)	(0.006)	(0.010)
Observations	591	633	647	895	895
R-squared				0.188	0.308
Control Mean	0.43	0.17	0.16	0.01	0.02
Team Controls	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes
Group × Period FE	Yes	Yes	Yes	Yes	Yes

Notes. This table reports estimates of the effect of the digital tracking system and its interaction with managerial quality on team-level outcomes. The sample excludes three teams that had high-performing managers before treatment but none afterward. The treatment indicator equals one if the product line associated with a team had implemented the system during the period. The managerial quality indicator equals one if the team had at least one high-performing manager at any point prior to being exposed to the system. The unit of observation is a team-bimonthly spell, and the sample covers the period from January 2020 to September 2024. Panel A presents output-related outcomes; Panel B reports human resource outcomes. Columns (1)–(8) in Panel B use Poisson regressions due to the count nature of the dependent variables; the remaining columns use OLS. In Poisson specifications, team-period observations with all-zero outcomes are dropped, resulting in smaller sample sizes. All regressions include team and group-specific period fixed effects, along with time-varying team-level covariates: the shares of workers who are female, ethnic Han, and college-educated, and the average age and tenure of both workers and managers. *Control Mean* reports the average of the dependent variable among untreated observations. Standard errors are clustered at the team level.

Table A18: The Effect of Losing a Good Manager

Variable	(1) IHS(Output per Worker)	(2) High Performers Transferred Out
Losing High-Performing Manager	-0.218	0.908
	(0.051)	(1.345)
Observations	564	296
R-squared	0.746	
Team Controls	Yes	Yes
Team FE	Yes	Yes
Period FE	Yes	Yes

Notes. This table reports estimates of the effect of losing high-performing managers on key team-level outcomes. The sample includes three teams that had high-performing managers prior to implementation but none afterward, as well as teams that never had any high-performing manager. The unit of observation is a team-bimonthly spell. Standard errors are clustered at the team level.

Table A19: Heterogeneous Effects by Managerial Quality (Continuous Measure)

Variable	(1) IHS(Output)	(2) IHS(Output per Worker)	(3) High Performers Transferred Out
Implemented	0.2814	-0.0498	0.6895
	(0.2594)	(0.0631)	(0.4090)
Implemented × Managerial Quality	0.1831	0.2057	-2.4900
	(0.5866)	(0.1095)	(0.6955)
Observations	955	955	642
R-squared	0.8189	0.6951	
Control Mean	12.28	9.83	0.43
Team Controls	Yes	Yes	Yes
Team FE	Yes	Yes	Yes
Group × Period FE	Yes	Yes	Yes

Notes. This table reports estimates of the effect of the digital tracking system and its interaction with managerial quality on key team-level outcomes. Managerial quality is based on the average pre-tracking leader allowance. The treatment indicator equals one if the product line associated with a team had implemented the system during the period. The continuous manager performance is defined as the highest mean leader allowance for a team before the implementation of the tracking system. The unit of observation is a team-bimonthly spell, and the sample covers the period from January 2020 to September 2024. Columns (1)-(2) are estimated using OLS, while column (3) uses a Poisson regression given the count nature of the dependent variable. In the Poisson specification, team-period observations with all-zero outcomes are dropped, leading to a smaller sample size. All regressions include team and period fixed effects, along with time-varying team-level covariates: the shares of workers who are female, ethnic Han, and college-educated, as well as the average age and tenure of both workers and managers. Control Mean reports the average of the dependent variable among untreated observations. Standard errors are clustered at the team level.

Table A20: Treatment Effects Holding Team Composition Fixed

VARIABLES	(1)	(2)	(3)	(4)
	CSPP		IHS(CSPP)	
Implemented	475.722	334.174	0.416	0.308
	(146.669)	(163.323)	(0.214)	(0.226)
Implemented × High-Performing Manager		411.168		0.336
		(318.323)		(0.389)
Observations	933	866	933	866
R-squared	0.784	0.719	0.886	0.458
Team Controls	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes
Period FE	Yes		Yes	
Group × Period FE		Yes		Yes

Notes. This table reports the effect of digital tracking on team performance when workforce composition is held fixed at each team's pre-implementation composition. The dependent variable is a composition-standardized measure of team-period performance pay constructed by classifying workers into three groups-old high performers, old low performers, and new entrants to the team—computing group-specific average performance pay within each team-period, and then aggregating these averages using team-specific weights based on the team's pre-implementation composition (composition standard performance pay, CSPP). Columns (1) and (2) use the standardized outcome in levels, and Columns (3) and (4) use its inverse hyperbolic sine (IHS). Columns (1) and (3) are estimated using Equation 5, and Columns (2) and (4) are estimated using Equation 6. The treatment indicator equals one once the team's product line has implemented the digital tracking system. The good-manager indicator equals one if the team had at least one manager whose average pre-tracking leader allowance ranked in the top 25% of the distribution. The unit of observation is a team-bimonthly period, and the sample covers January 2020 to September 2024. All regressions include team and period fixed effects, as well as time-varying team-level controls: the share of workers who are female, ethnic Han, and college-educated, and the average age and tenure of both workers and managers. Standard errors are clustered at the team level.

Table A21: Effects of Managerial Quality on Workers' Outcomes

Panel A. Execution Errors

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Making Mistakes							
	OLS				Probit			
	All Workers	High Performers	Low Performers	New Workers	All Workers	High Performers	Low Performers	New Workers
Supervised by a High-Performing Manager	-0.095	-0.110	-0.005	-0.318	-0.319	-0.348	-0.055	-1.030
	(0.029)	(0.036)	(0.058)	(0.118)	(0.066)	(0.075)	(0.186)	(0.675)
Observations	12,962	8,169	2,748	1,920	12,494	7,753	2,047	1,537
R-squared	0.317	0.328	0.336	0.481				
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Panel B. Absenteeism

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	IHS(Absent Rate)				Absent Rate			
	OLS				OLS			
	All Workers	High Performers	Low Performers	New Workers	All Workers	High Performers	Low Performers	New Workers
Supervised by a High-Performing Manager	0.000	-0.012	-0.006	0.041	-0.000	-0.014	-0.007	0.044
	(0.005)	(0.003)	(0.009)	(0.017)	(0.005)	(0.003)	(0.009)	(0.018)
Observations	14,607	6,371	3,220	5,016	14,607	6,371	3,220	5,016
R-squared	0.248	0.112	0.255	0.289	0.249	0.113	0.254	0.293
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. This table reports the effects of managerial quality on workers' outcomes related to burnout during the period. The key independent variable is a time-varying indicator for whether the worker was supervised by a high-performing manager. Panel A examines execution errors, where the dependent variable is a binary indicator for whether the worker made at least one mistake on a given day. Columns (1)–(4) are estimated using OLS; Columns (5)–(8) use Probit models. The unit of observation is a worker-day. Panel B focuses on absenteeism, where the dependent variable is the worker's monthly absence rate. Columns (1)–(4) use the inverse hyperbolic sine (IHS) transformation of the absence rate, while Columns (5)–(8) use the raw rate. All columns are estimated using OLS. The unit of observation is a worker-month. The unit of observation is a worker-day in Panel A and worker-month in Panel B. In both panels, Columns (1) and (5) include all workers; Columns (2) and (6) focus on high performers; Columns (3) and (7) on low performers; and Columns (4) and (8) on new hires who joined after the system implementation. All regressions include worker and date fixed effects (month fixed effects in Panel B) and cluster standard errors at the worker level.

Table A22: High Performers Self-reported Overload with Different Managers

Variable	(1) Overload	(2) Overload	(3) Overload
Supervised by a High-Performing Manager	-0.783	-0.782	-0.741
	(0.415)	(0.419)	(0.404)
Observations	266	266	264
Performance Wage Controlled		Yes	Yes
Demographics Controlled			Yes

Notes. This table reports cross-sectional regressions estimating the effect of supervision by a high-performing manager on the likelihood that high-performing workers report feeling overloaded (agree or highly agree to the statement "I feel my workload is too heavy") in September 2024. Controls include individual performance pay and demographic characteristics. Standard errors are clustered at the team level.

Table A23: Alternative Mechanisms: Payment and Promotion

Variable	(1) IHS(Performance Wage)	(2) IHS(Performance Wage)	(3) Promotion	(4) Promotion
	High Performers	Low Performers	High Performers	Low Performers
Implemented	0.028	0.154	1.233	0.642
	(0.031)	(0.046)	(0.851)	(0.487)
Implemented × High-Performing Manager	0.057	0.230	-2.045	0.841
	(0.057)	(0.119)	(2.401)	(0.770)
High-Performing Manager	-0.031	-0.218	2.267	-0.744
	(0.038)	(0.103)	(2.764)	(0.796)
Observations	17,961	8,473	17,975	8,479
R-squared	0.464	0.473	0.048	0.111
Worker FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes

Notes. This table reports whether the effects of digital tracking differ by managerial quality for worker compensation and promotion outcomes. The dependent variables in Columns (1) and (2) are the inverse hyperbolic sine (IHS) of monthly performance pay, and the dependent variables in Columns (3) and (4) are indicators for promotion in a given month. Columns (1) and (3) restrict the sample to high-performing workers, and Columns (2) and (4) restrict it to low-performing workers. The key regressors are indicators for system implementation, supervision by a high-performing manager, and their interaction. The unit of observation is worker-month, covering the period from January 2020 to July 2024. All regressions include worker and month fixed effects. Standard errors are clustered at the worker level.

Table A24: Alternative Explanations: System Adoption and Material Shortage

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	No Auto-termination				No Shortage			
	All Workers	High Performers	Low Performers	New Workers	All Workers	High Performers	Low Performers	New Workers
Supervised by a High-Performing Manager	0.051	0.059	0.016	0.213	0.036	0.034	0.039	-0.105
	(0.019)	(0.022)	(0.033)	(0.161)	(0.039)	(0.042)	(0.126)	(0.080)
Observations	12,962	8,169	2,748	1,920	12,962	8,169	2,748	1,920
R-squared	0.439	0.435	0.434	0.542	0.379	0.385	0.460	0.514
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. This table reports estimates of the effect of managerial quality on workers' compliance with the digital tracking system and incidence of material shortages. The key independent variable is a dummy indicating whether the worker was supervised by a high-performing manager on a given day. The dependent variable in Columns (1)–(4) is an indicator for having no auto-terminated tasks that day, while Columns (5)–(8) use an indicator for experiencing no material shortages. Some observations are dropped in grouped regressions due to becoming singleton units when including date fixed effects. The sample covers January 2022 to July 2024. All regressions include worker and date fixed effects, with standard errors clustered at the worker level.

Table A25: Decomposing Productivity Gains

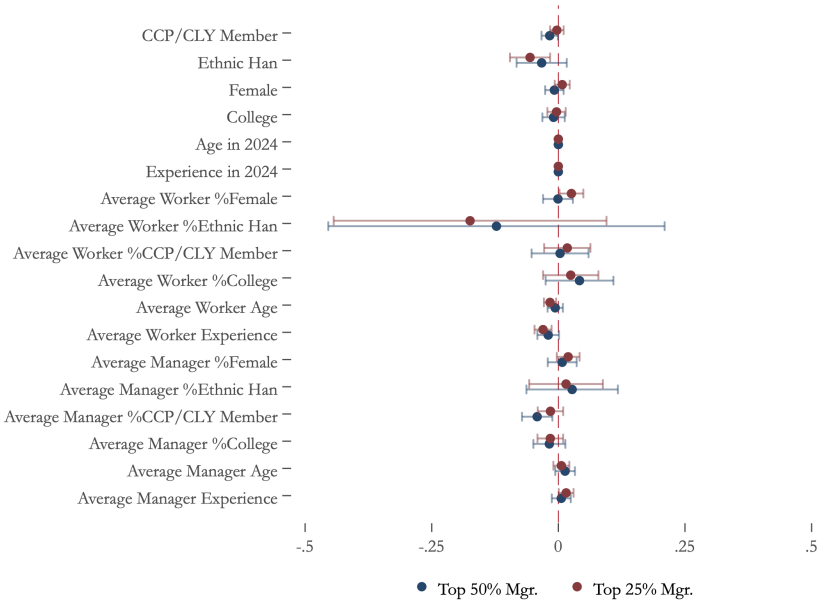
Variable	(1)	(2)	(3)	(4)
	Output per Worker (High)		Output per Worker (Low)	
	raw	IHS	raw	IHS
Implemented	639.703	0.064	85.197	0.039
	(501.300)	(0.042)	(406.136)	(0.058)
Implemented × High-Performing Manager	1,264.663	0.063	2,659.784	0.249
	(1,153.568)	(0.070)	(1,225.080)	(0.106)
Observations	950	950	903	903
R-squared	0.702	0.699	0.698	0.658
Team Controls	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes
Group×Period FE	Yes	Yes	Yes	Yes

Notes. This table reports estimates of the effect of the digital tracking system and its interaction with managerial quality on output per worker, separately for high-performing workers and low-performers. The dependent variables are output per worker type—that is, total output by high-performing workers (or low-performers) divided by their respective counts. The unit of observation is a team-bimonthly period. All regressions control for team and group-specific period fixed effects, as well as time-varying team characteristics. Standard errors are clustered at the team level.

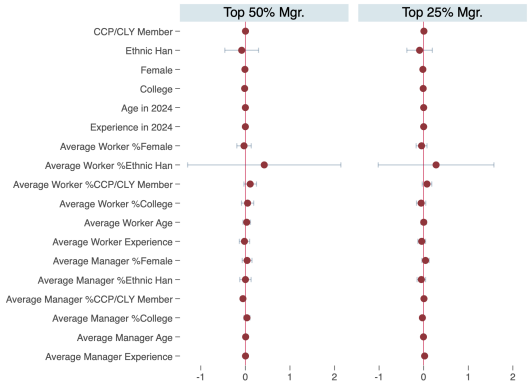
Table A26: Other Alternative Explanations for High Performers Retention

	(1)	(2)	(3)	(4)
	Benefits and Opportunities		Perceived Manager Ability	
Variable	Promotion Chance	Benefits Provided	Good Scheduling	Good Cooperation
High Performer	-0.182	-0.116	-0.154	-0.058
	(0.144)	(0.104)	(0.111)	(0.081)
High-Performing Manager × High Performer	-0.060	-0.032	-0.028	-0.139
	(0.210)	(0.145)	(0.168)	(0.144)
Observations	578	578	578	578
R-squared	0.084	0.086	0.048	0.050
Team FE	Yes	Yes	Yes	Yes
	(5)	(6)	(7)	(8)
	Job Meaning		Communication Quality	
Variable	Importance for Company	Importance for Society	Communication Times	Communication Smoothness
High Performer	-0.059	0.044	0.765	-0.063
	(0.091)	(0.087)	(0.849)	(0.124)
High-Performing Manager × High Performer	0.191	-0.009	-0.625	-0.253
	(0.180)	(0.171)	(1.336)	(0.168)
Observations	578	578	578	578
R-squared	0.093	0.073	0.087	0.103
Team FE	Yes	Yes	Yes	Yes

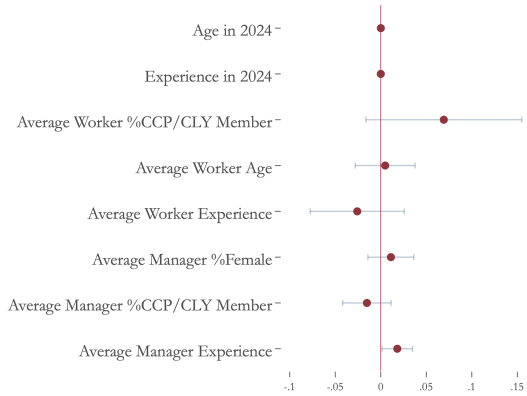
Notes. This table reports estimates of managerial quality effects on various worker-reported outcomes from the September 2024 survey, including satisfaction with current benefits and opportunities, perceptions of team managers' scheduling ability and cooperation skills, beliefs about the job's importance to the company and society, and daily communication frequency and smoothness with managers. The key independent variables are a high performer indicator and its interaction with a dummy for supervision by a high-performing manager. The sample includes all surveyed workers (excluding managers), with each observation representing a worker. All regressions control for team fixed effects. Standard errors are clustered at the team level.



(a) Univariate Regressions



(b) Joint Regression



(c) Lasso Selected Variables (Top 25%)

Figure A1: Factors associated with Managerial Quality

Notes. This figure presents the tests for the prediction power of manager characteristics for whether a manager is classified as high-performing, which is defined using two criteria: those whose average leader allowance falls in the top 50% and top 25% of the distribution. Panel (a) reports coefficient estimates and 95% confidence intervals from separate (univariate) regressions that include each predictor one at a time. Panel (b) reports estimates from a single regression that includes the full set of predictors simultaneously. Panel (c) reports estimates from a post-LASSO specification: predictors are first selected via cross-validated LASSO logit, and the outcome is then regressed on the selected predictors. Cross-validated LASSO logit selects $\lambda = 0.1618$ with 0 nonzero coefficients for top 50% managers, and $\lambda = 0.0403$ with 8 nonzero coefficients for top 25% managers. The regressors include basic demographic characteristics of the managers themselves, the average demographic characteristics of workers under their supervision, and the characteristics of other managers supervising the same team. All demographic variables have been multiplied by 10 for clearer presentation of the survey variables' effects.

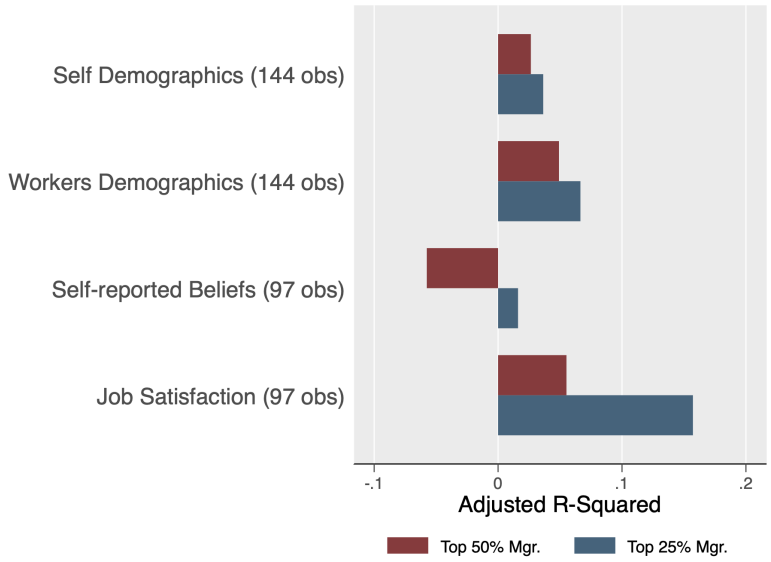


Figure A2: Predictive Power of Observable Factors for Managerial Quality

Notes. This figure presents how well different factors explain whether a manager is classified as high-performing under different criteria, measured by the cumulative adjusted R^2 of regression models. The red bars use an indicator for whether a manager's mean leader allowance before implementation lies in the top 50% of the distribution, while the blue bars use the corresponding top-25% indicator. This graph tracks how the R^2 changes as four groups of variables are gradually added: self demographics, workers' demographics, self-reported beliefs, and job satisfaction. A negative adjusted R^2 is entirely normal in this setting and simply indicates that, after accounting for model complexity, the covariates provide essentially no predictive power beyond an intercept-only benchmark.

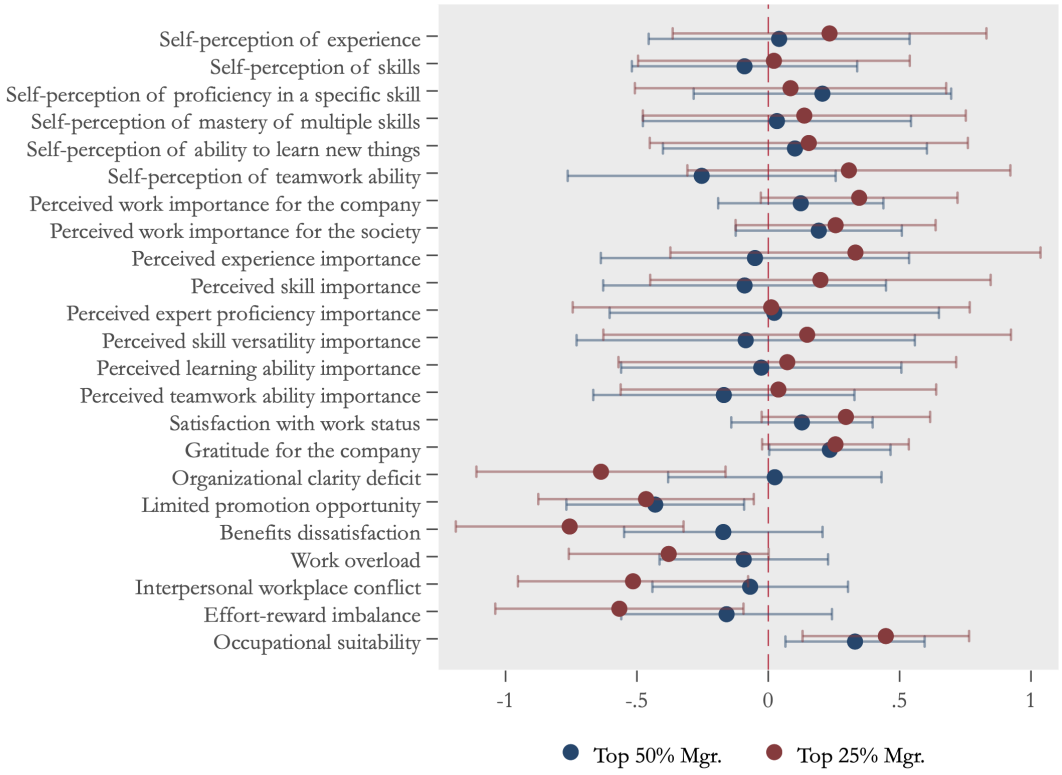


Figure A3: Difference in Survey Responses by Manager Types

Notes. This figure presents the coefficients and 95% confidence intervals of regressions of survey responses on the indicator for whether the manager is classified as high-performing under two criteria: those whose average leader allowance falls in the top 50% (blue) and top 25% (red) of the distribution. The response values range from 1 to 4, with 4 as mostly agree and 1 as mostly disagree.

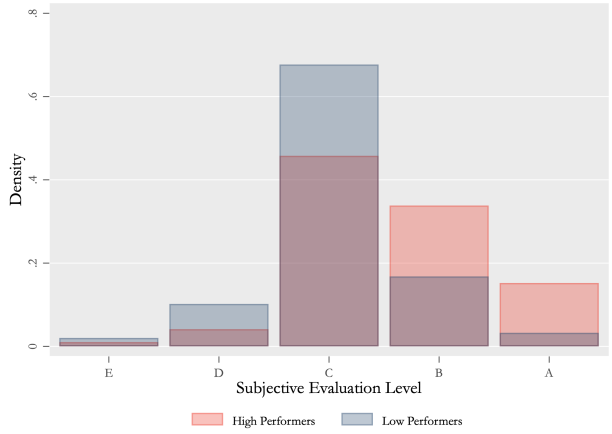
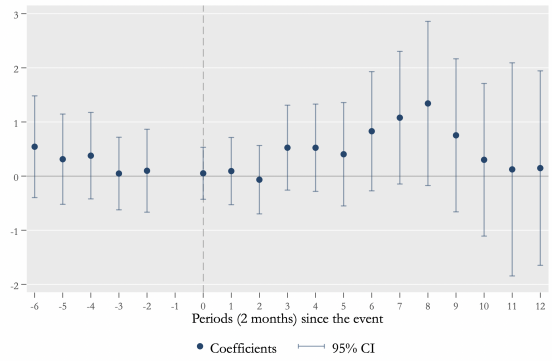
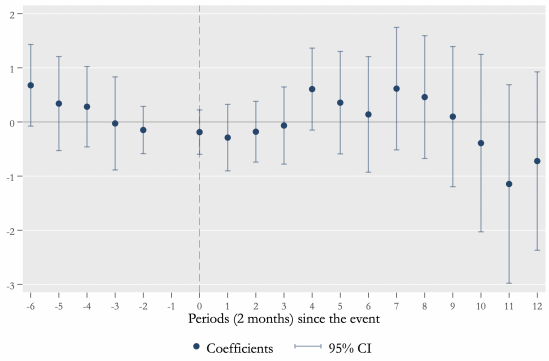


Figure A4: Subjective Evaluation Distribution of High and Low Performers

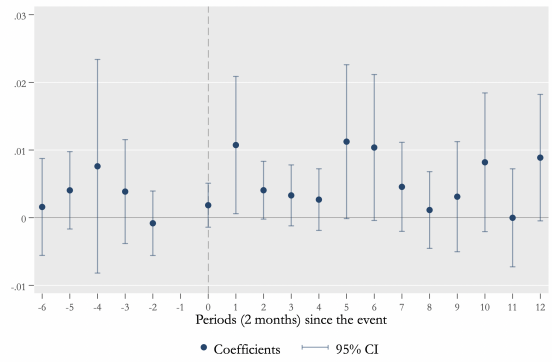
Notes. This figure presents the subjective evaluation level distribution of high performers and low performers. Each observation is a worker-month and the time span is from January 2023 to September 2024. There are 5 levels, with A as the best and E as the worst rating.



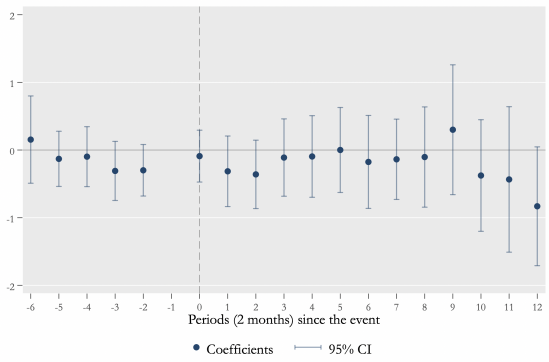
(a) High Performers Transferred In



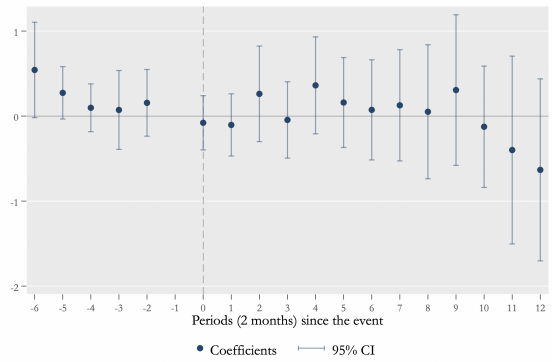
(b) High Performers Transferred Out



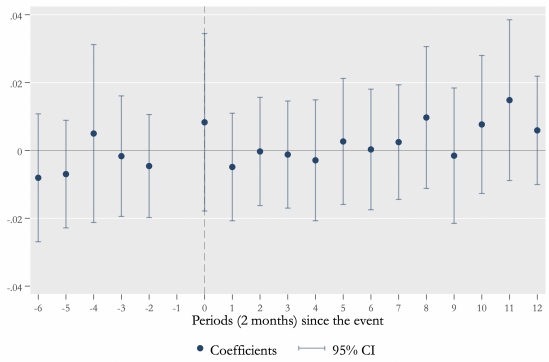
(c) High Performer Turnover Rate



(d) Low Performers Transferred In



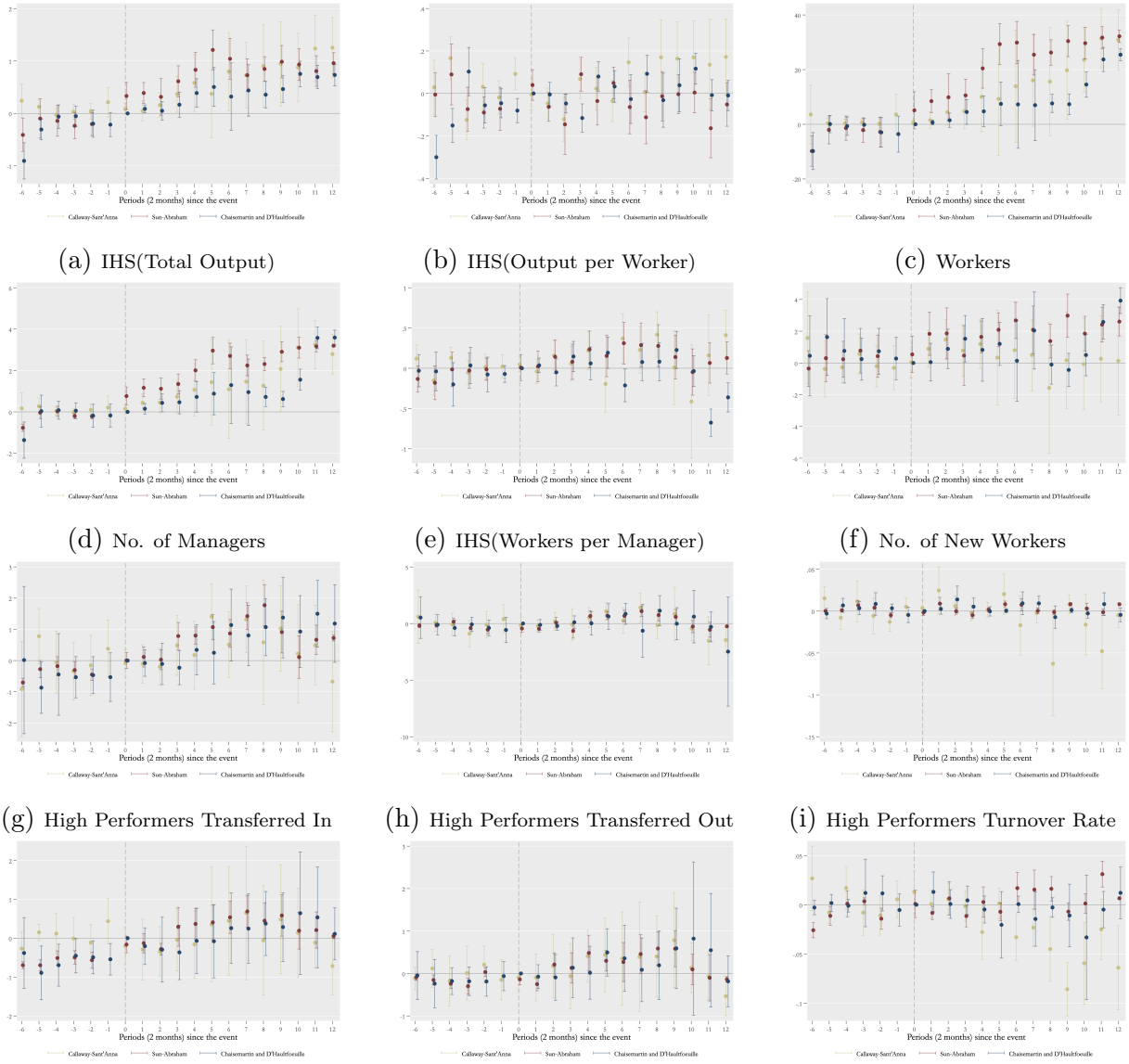
(e) Low Performers Transferred Out



(f) Low Performer Turnover Rate

Figure A5: Team-Level Event Studies (Personnel Transfers)

Notes. This figure presents event-study estimates and 95% confidence intervals from equation (4) for personnel-transfer outcomes. Treatment is defined as exposure of a team's product line to the digital tracking system. The vertical dashed line indicates treatment timing. Control teams are those not yet treated. The sample is a team-bimonthly panel of 38 teams from January 2020 to September 2024.



(a) IHS(Total Output) (b) IHS(Output per Worker) (c) Workers

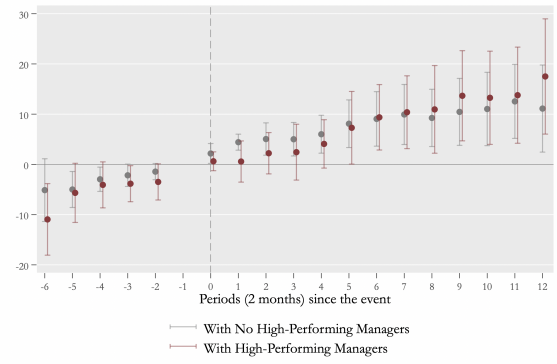
(d) No. of Managers (e) IHS(Workers per Manager) (f) No. of New Workers

(g) High Performers Transferred In (h) High Performers Transferred Out (i) High Performers Turnover Rate

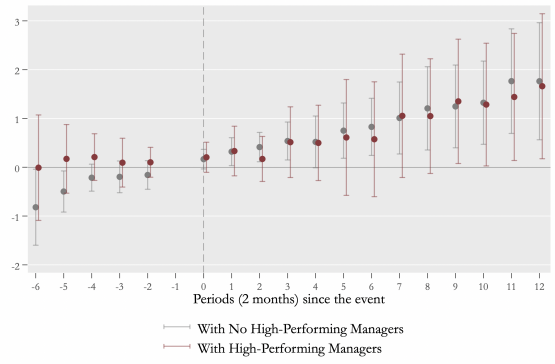
(j) Low Performers Transferred In (k) Low Performers Transferred Out (l) Low Performers Turnover Rate

Figure A6: DiD Corrections for Event Studies on Main Outcomes (Team Level)

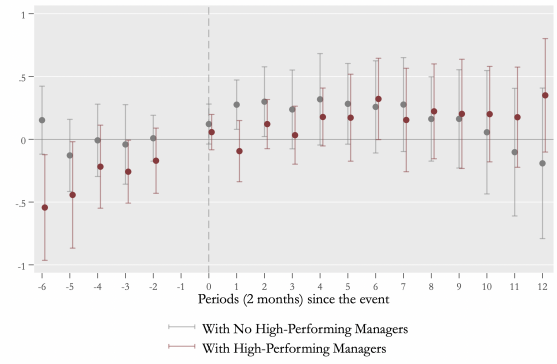
Notes. This figure presents robustness checks for the event-study results shown in Figure 2, based on the estimators proposed by Callaway and Sant'Anna (2021), Sun and Abraham (2021), and De Chaisemartin and d'Haultfoeuille (2020). Treatment is defined as exposure of a team's product line to the digital tracking system. The vertical dashed line indicates treatment timing. Control teams are those not yet treated or never treated. The sample is a team-bimonthly panel of 38 teams from January 2020 to September 2024.



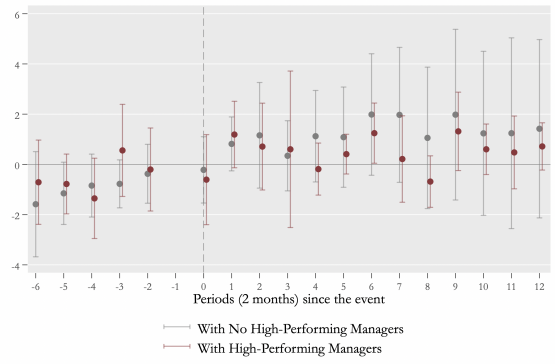
(a) Team Size



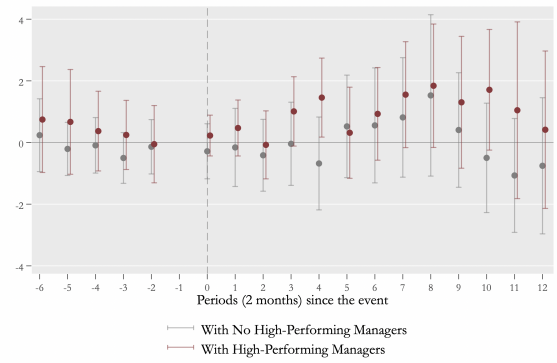
(b) No. of Managers



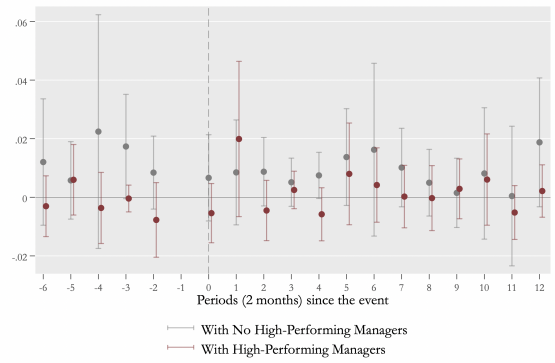
(c) IHS(Workers per Manager)



(d) No. of New Workers



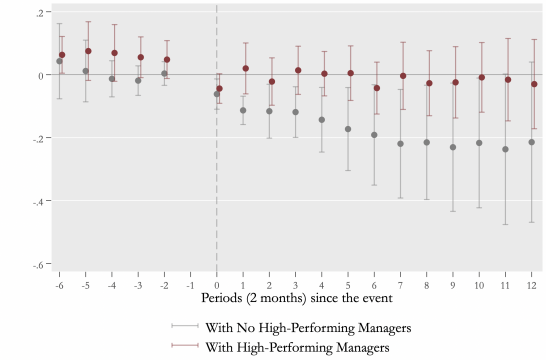
(e) High Performers Transferred In



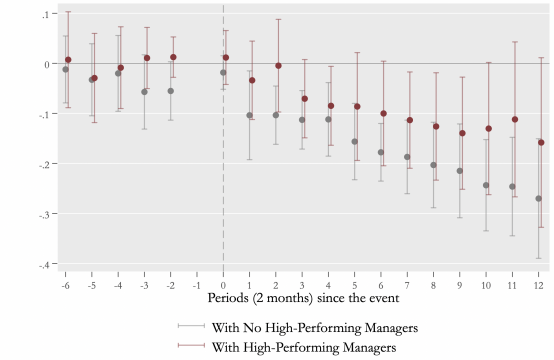
(f) High Performer Turnover Rate

Figure A7: Team-Level Heterogeneous Event Studies

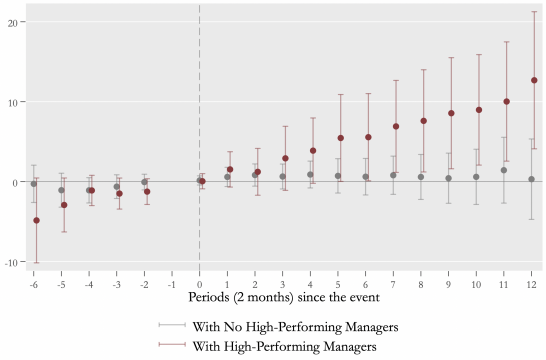
Notes. This figure presents grouped event-study estimates and 95% confidence intervals from equation (4) on additional outcome variables, estimated separately for teams with high-performing managers and others. Red markers denote teams with high-performing managers, and gray markers denote other teams. Treatment is defined as exposure of a team's product line to the digital tracking system. The vertical dashed line indicates treatment timing. Control teams are those not yet treated. The sample is a team-bimonthly panel of 38 teams from January 2020 to September 2024.



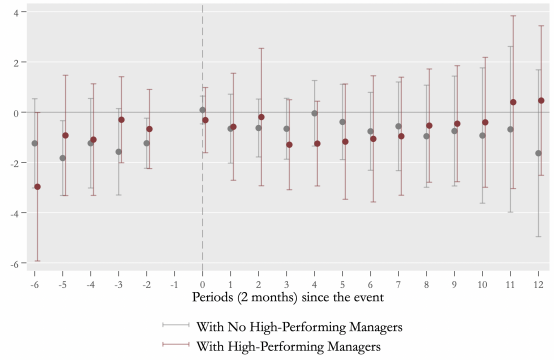
(a) IHS(High Performers Share)



(b) IHS(Low Performers Share)



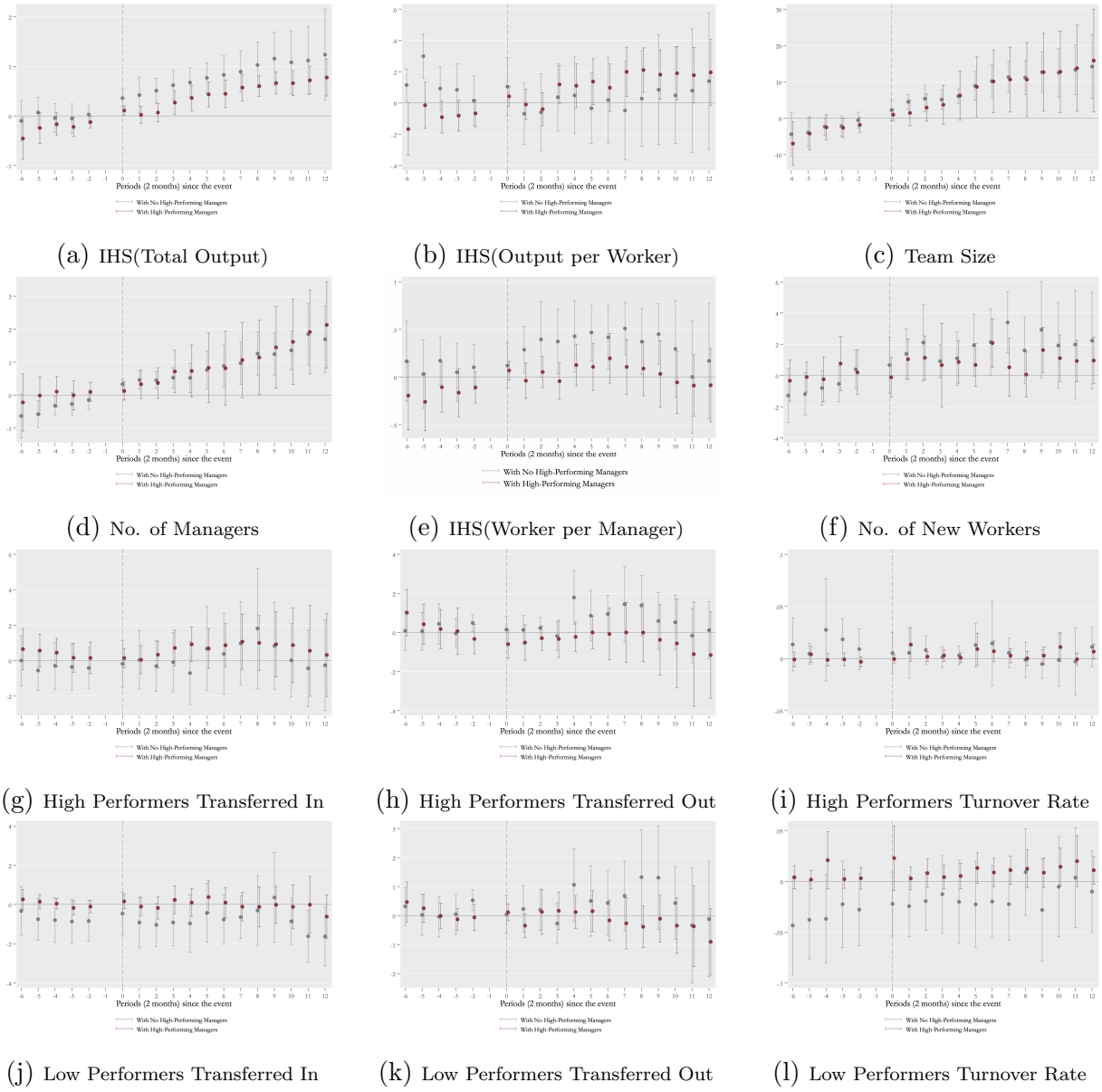
(c) No. of High Performers



(d) No. of Low Performers

Figure A8: Team-Level Heterogeneous Event Studies (Team Composition)

Notes. This figure presents grouped event-study estimates and 95% confidence intervals from equation (4) on the shares and counts of different types of workers for teams with high-performing managers and others. Red markers denote teams with high-performing managers, and gray markers denote other teams. Treatment is defined as exposure of a team's product line to the digital tracking system. The vertical dashed line indicates treatment timing. Control teams are those not yet treated. The sample is a team-bimonthly panel of 38 teams from January 2020 to September 2024.



(a) IHS(Total Output) (b) IHS(Output per Worker) (c) Team Size

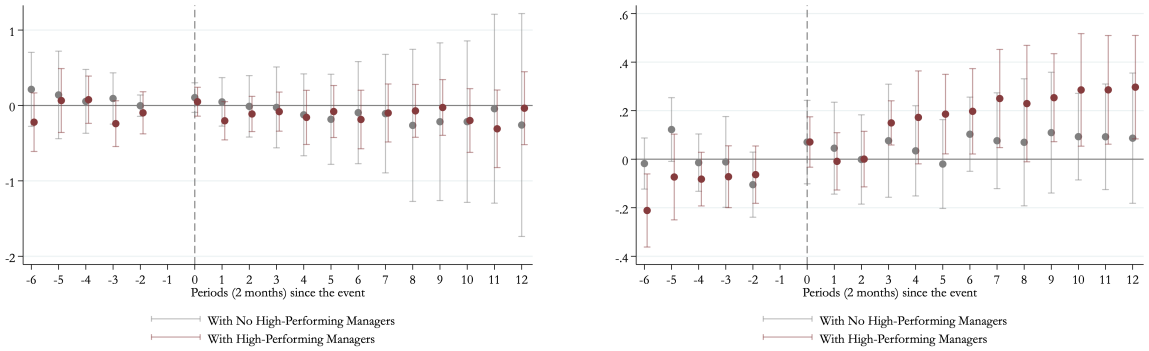
(d) No. of Managers (e) IHS(Worker per Manager) (f) No. of New Workers

(g) High Performers Transferred In (h) High Performers Transferred Out (i) High Performers Turnover Rate

(j) Low Performers Transferred In (k) Low Performers Transferred Out (l) Low Performers Turnover Rate

Figure A9: Heterogeneous Event Studies (50%)

Notes. This figure presents grouped event-study estimates and 95% confidence intervals from equation (4), estimated separately for teams with high-performing managers and other teams, where high-performing managers are defined as those in the top 50% of the pre-tracking leader-allowance distribution. Red markers denote teams with high-performing managers, and gray markers denote other teams. Treatment is defined as exposure of a team's product line to the digital tracking system. The vertical dashed line indicates treatment timing. Control teams are those not yet treated. The sample is a team-bimonthly panel of 38 teams from January 2020 to September 2024.



(a) IHS(Total Output) (b) IHS(Output per Worker)

Figure A10: Heterogeneous Event Studies (Only Workers in the Teams Pre-Implementation)

Notes. This figure presents grouped event-study estimates and 95% confidence intervals from equation (4) for total output and output per worker, estimated separately for teams with high-performing managers and other teams. The sample is restricted to workers who were already in the team before implementation. Red markers denote teams with high-performing managers, and gray markers denote other teams. Treatment is defined as exposure of a team's product line to the digital tracking system. The vertical dashed line indicates treatment timing. Control teams are those not yet treated. The sample is a team-bimonthly panel of 38 teams from January 2020 to September 2024.